

Review Article

A Review of Research on Flight Delay Propagation: Current Situation and Prospect

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The correlation between flight routes makes the initial delays easy to cause chain delays of downstream airports, forming a large area of delays. Studying delay propagation can block its propagation path and alleviate large-scale delay problems. This paper first systematically collects relevant academic literature from the past decade and constructs a co-occurrence keyword network to analyze and reveal the research hotspots in flight delay propagation. Then, based on the two major categories of delay propagation within airlines and between airports, this paper provides a detailed review and summary of their research methods. For the research of delay propagation within airlines, scholars mainly use econometric models, Bayesian networks, function models, and propagation trees to analyze the influencing factors and propagation characteristics of delay propagation. Among the methods for predicting delay propagation, models based on machine learning algorithms account for a large proportion and have shown good prediction performance. For the more complex delay propagation problem between airports, researchers mainly use the time interval Petri net, queuing network model, Cox proportional hazards model, and complex network theory to analyze the delay propagation mechanism in aviation networks. In addition, deep learning models and spatiotemporal network models have improved the accuracy of interairport flight delay prediction due to their ability to process large datasets and high-dimensional feature data. Finally, this paper summarizes the progress and shortcomings in flight delay propagation. The results show that there are significant differences in the delay propagation mechanism between airlines and airports, which requires full consideration of their applicability when selecting predictive models. Traditional machine learning methods perform well in the internal delay prediction of airlines, but there are some limitations in dealing with the complex and changeable propagation environment between airports. On the contrary, deep learning models and spatiotemporal models have opened up a new path for improving prediction accuracy by their powerful data processing and analysis capabilities. At the same time, researchers also need to constantly explore and optimize algorithms to overcome their current limitations and further improve the reliability of predictions.

Keywords: delay propagation; flight delay; prediction methods; propagation mechanisms

1. Introduction

With the gradual control of the COVID-19 epidemic and the recovery of the social economy, the global air transport industry has ushered in a new stage of rapid recovery and development. However, in the context of the surge in aviation market demand and the increasing density of flights, the problem of flight delays has once again become the focus of widespread concern in the air transport industry [1, 2].

Flight delays not only affect the travel experience of passengers but also pose great challenges to airline operational efficiency, airport resource allocation, and air traffic management [3–5].

The U.S. Federal Aviation Administration defines flights that experience delays exceeding 15 min as delayed flights [6]. There are many factors that can cause flight delays, including weather [7], airlines [8], flight schedules [9], airports [10], and air traffic control [11]. Therefore, flight

delays are inevitable. At the same time, in the aviation network, flight delays will generate delay propagation phenomena through the network as a carrier [12]; that is, an aircraft will execute multiple flights every day, and the flight schedule is relatively tight, with little room for time reversal. Therefore, after a flight is delayed, it will greatly affect the supply and demand of flight resources for subsequent flights, causing a chain reaction of subsequent flights and leading to the occurrence of delay propagation [13]. Therefore, flight delay propagation can be defined as the phenomenon of delays in other flights in the aviation network caused by initial flight delays [14]. Flight delay propagation is mainly caused by factors such as flight scheduling, flight resource sharing, crew scheduling, airport operations, climate change, traffic control, and passenger reasons [15].

Flight delay propagation can be divided into two categories: delay propagation within airlines and delay propagation between airports. The delay propagation within airlines is mainly influenced by the close correlation between crew deployment, aircraft rotation, and flight schedules. Due to the prefixed flight chain of aircraft, there is a certain hidden correlation between multiple flights, and the delay of some flights will trigger a chain reaction [16], which will spread to other flights and have a significant impact on the delay performance of downstream aircraft [17], leading to further delays in flight schedules [18]. The delay propagation between airports focuses more on the connectivity characteristics of airport nodes and routes in the aviation network, where many flights gather at the airport and wait for take-off and landing in a specified order. Due to the exclusivity of parking spaces and runways in use [19], the arrival and departure of delayed flights will disrupt the airport schedule and lead to an imbalance in the supply and demand of various flight resources, including airport capacity and airflow [20]. Therefore, airport capacity, weather conditions, and route congestion often become key factors in the propagation of delays between different airports. These two types of delay propagation mechanisms have both independent characteristics and are intertwined due to the linkage of aviation networks, forming complex propagation paths. The flight delay propagation not only puts enormous pressure on airline operations and airport scheduling [21] but also may lead to passenger travel disruptions [22], increased economic losses [23], and increased psychological stress [24]. The chain reaction of delays may lead to a backlog of subsequent flights, overload of airport resources, and even trigger widespread flight cancellations.

Therefore, studying the propagation law of flight delays in the aviation network can provide a theoretical basis for government departments to control flight delays. It can reveal the delay propagation mechanism and key influencing factors. By analyzing the influence of various factors on the propagation of flight delays, effective prevention and control methods can be found, which is of great significance to improve the management level of flight delays [25]. In addition, studying the propagation of flight delays can help airlines and air transport management departments to optimize flight planning and resource allocation [26]. By predicting the possibility of delay propagation, airlines and

air transport management departments can take measures in advance, such as adjusting crew schedules and arranging backup aircraft reasonably, thereby reducing operating costs, improving service quality, and reducing customer complaints and brand losses caused by delays [27]. For airports, studying flight delay propagation can help optimize the scheduling and utilization of ground resources, such as parking stands, boarding gates, and baggage handling equipment [28]. Understanding the propagation pattern of delays can help airports develop emergency plans, improve operational efficiency, alleviate congestion caused by dense flights, and ensure the overall smooth and orderly operation of the airport. In the air traffic management, studying flight delay propagation can help optimize airspace allocation and flow control, and reduce route congestion and the possibility of secondary delays [29]. By analyzing the propagation path of delays, air traffic control departments can develop more accurate traffic management strategies, improve airspace operational efficiency, ensure flight safety, and provide a more stable operating environment for the entire air transport system [30].

The purpose of this paper is to systematically review the related research on flight delay propagation and explore the core mechanism, prediction methods, and application potential of different propagation types. This paper first collects literature from the past decade for network analysis and divides the literature into two categories based on the different research objects: delay propagation within airlines and delay propagation between airports. Through co-occurrence keyword network analysis, it is found that the hotspots in the field of flight delay include the construction of delay models, model optimization, delay propagation, and air traffic management. In flight delay prediction, machine learning, deep learning, and air traffic flow management (ATFM) methods are particularly prominent. Subsequently, this paper compares and analyzes the similarities and differences between two types of delay propagation mechanisms and the methods for predicting delay propagation. Research has found that traditional machine learning methods perform well in predicting delay propagation within airlines, but have certain limitations in dealing with complex and changing communication environments between airports. The prediction of delay propagation between airports mainly uses methods such as machine learning, deep learning, spatiotemporal network modeling, and ATFM methods. Finally, this paper summarizes the current research progress and future development directions. Through this review, it can provide research references for further exploring the mechanism and prediction methods of flight delay propagation and provide theoretical support for optimizing the management of air transport systems. It is of great significance for reducing the impact of flight delays, saving time, improving transportation efficiency, and reducing transportation costs.

2. Materials and Methods

This paper focuses on the hot spots of flight delays based on the Web of Science Core Collection Database (SCI-

EXPANDED). This paper selects the relevant literature from 2014 to 2024, establishes a co-occurrence keyword network, and uses Cite Space software to analyze the network.

2.1. Data Sources and Research Methods. The research data related to this paper are from the Web of Science Core Collection Database (SCI-EXPANDED). The search Equation is as follows: (TS=(Flight delay propagation prediction)) OR TS=(Flight delay prediction)) OR TS=(Flight delay). The time is selected from 2014 to 2024. The type of publication is limited to Article or Article Review, and the language is limited to English. In order to ensure the accuracy of data sources, this paper finally selects 641 valid documents. Since 2014, the number of published papers has continued to increase overall.

2.2. Co-Occurrence Keyword Knowledge Graph. The co-occurrence keyword network is a network structure constructed by analyzing the co-occurrence relationship between keywords in the text data, which can show the keywords of the selected article and the relationship between the keywords. Several keywords in a paper will have a specific correlation, which can be expressed by co-occurrence frequency. Drawing the keyword co-occurrence network through Cite Space software can analyze the frequency and trend of keywords in specific fields or topics, reveal the research hotspots and development trends in this field, and provide valuable reference information for researchers.

The keyword co-occurrence network in flight delays is shown in Figure 1. The network has 314 nodes and 1872 connections. The network density is 0.0381. Among them, the nodes of model, optimization, air traffic management, aircraft, delay propagation, and other keywords are the largest, indicating that these keywords appear frequently and are hot topics of academic concern. The betweenness centrality of model keywords is more significant than 0.1, indicating that it is a pivotal bridge in linking scholarly works across diverse research topics and disciplines. It is the critical research path and turning point in the research field, highlighting the importance of the node in the network structure. In addition, the nodes of the network, delay, impact, machine learning, flight delays, and other keywords are also prominent, showing that they also have a certain degree of attention in this field. According to the keyword co-occurrence map, the top 10 keywords are shown in Table 1, which can clearly show the hot research topics in flight delays.

2.3. Community Analysis of Co-Occurrence Keyword Network. To further explore the hot spots in flight delay research, this paper uses Cite Space software to extract cluster labels, and the obtained cluster labels can reflect the research situation. Through the keyword network clustering of the above 641 articles, as shown in Figure 2, seven clusters are finally obtained, showing seven hot topics in flight delay research. The first category is flight delay prediction, including deep learning, flight delay, machine learning, and ATFM. The

second category is column generation, including disruption management, aircraft routing, airline disruption management, and airline recovery. The third category is ATFM, including air traffic control, air traffic management, speed control, and linear holding. The fourth category is delay propagation, including resilience, complex networks, air transport networks, and Granger causality. The fifth category is air traffic management, including aircraft sequencing and scheduling problems, artificial intelligence, genetic algorithms, and delay propagation. The sixth category is atmospheric modeling, including the aviation industry, delays, prediction models, and airports. The seventh category is dynamic programming, including benefits of integration, airport congestion mitigation, airport demand management, and queuing model. In the map, the S value is $0.7475 > 0.7$, and the Q value is $0.3825 > 0.3$, indicating that the clustering map is reasonable and the clustering effect is convincing.

In addition, the timeline chart can display the time periods and trends of different keywords within each cluster, as well as the changes in research popularity of each topic over different years. As shown in Figure 3, flight delay prediction, column generation, ATFM, and delay propagation are the most prolonged hot research fields. In contrast, the research of model, delay propagation, delay, and other keywords has continued from 2014 to the present.

The research on flight delay propagation is clustered into 7 core groups around the main thread of “prediction—management—optimization.” The clustering “column generation” and “dynamic programming” emerged mainly in 2014–2018, while nodes such as “integer programming” and “resource allocation” were concentrated around 2016. Such methods are mostly used to solve combinatorial optimization problems in flight scheduling and lay the algorithmic foundation for subsequent research on delay propagation. The clustering “ATFM” and “air traffic management” also began to take shape, reflecting early explorations of macroscopic traffic control. From 2018 to 2021, clustering “flight delay prediction” and “delay propagation” became the research focus, and the research shifted to data-driven delay prediction and propagation analysis, with keywords such as “machine learning” and “complex network propagation” showing a significant increase in frequency. Support vector machines (SVM) and random forest algorithms are widely applied in delay prediction. During the same period, clustering “atmospheric modeling” began to emerge, with nodes such as “meteorological coupling analysis” and “extreme weather delay” marking the shift of research from a single human factor to multisource risk coupling. From 2021 to 2024, there was a trend of intelligent and systematic integration, with clustering “ATFM” and “air traffic management” demonstrating breakthroughs in real-time dynamic control technologies, achieving deep integration of meteorological data and traffic management. In addition, the application of reinforcement learning and deep neural networks in clustering “flight delay prediction” has significantly improved prediction accuracy and has formed a cross-clustering technology closed loop with “column generation” through the “online optimization” node, highlighting the integrated research paradigm of “prediction–decision”.

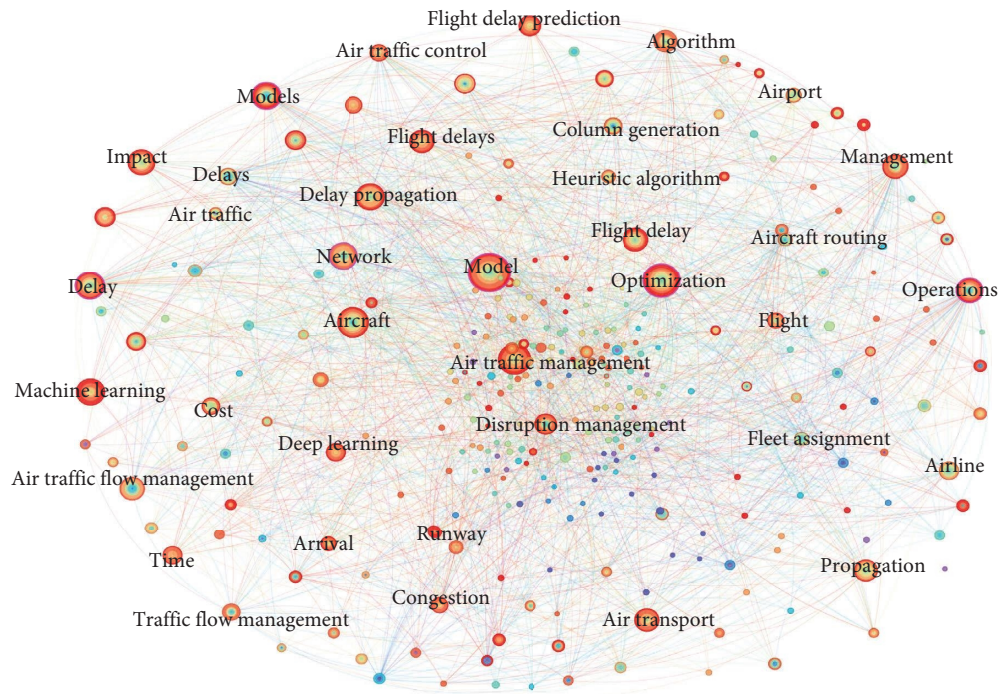


FIGURE 1: Knowledge graph of co-occurrence keywords.

TABLE 1: Top 10 keywords of word frequency.

Frequency	Centrality	Year	Keyword
87	0.16	2014	Model
71	0.1	2014	Optimization
65	0.05	2014	Air traffic management
55	0.09	2014	Aircraft
46	0.09	2014	Delay propagation
43	0.13	2014	Network
41	0.14	2014	Delay
41	0.02	2016	Impact
39	0.01	2019	Machine learning
37	0.04	2014	Flight delays

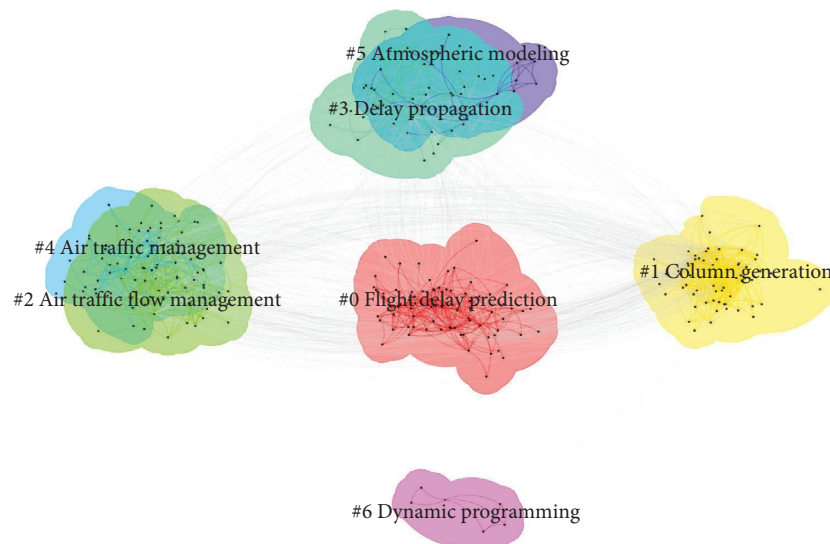


FIGURE 2: Keyword clustering map.

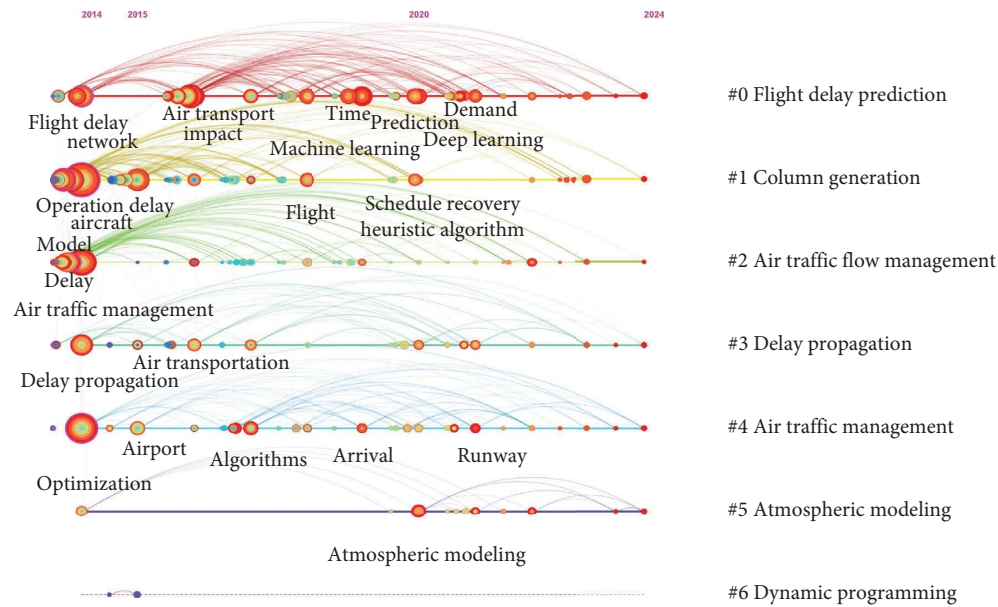


FIGURE 3: Keyword timeline diagram.

2.4. Statistical Analysis Summary. According to research and practice in the field of air transport, there are currently two main areas of delay propagation: One is the propagation of flight delays within airlines due to the interdependence between flights, and the other is the propagation of flight delays between airports due to the mutual influence and occupation of airport resources between flights. Through the statistical analysis of the above research results in flight delays, it can be seen that flight delay prediction has always been a hot spot. With the increasing demand for flights, the propagation of delays in the airport network is becoming more and more serious, and there are more and more studies on delay propagation.

When studying the propagation of flight delays, the prediction of flight delay propagation and the analysis of flight delay propagation characteristics are complementary links. The analysis of flight delay propagation characteristics reveals the occurrence patterns and propagation mechanisms of delay propagation through mining historical data, providing a basis for identifying key influencing factors and constructing prediction models. The prediction of flight delay propagation utilizes the analyzed patterns and models to estimate the future propagation of flight delays, helping to develop more efficient operational strategies and management decisions.

In addition, keyword analysis shows that the current hot topics in the field of flight delays include the construction of delay models, model optimization, delay propagation, air traffic management, etc. In flight delay prediction, machine learning, deep learning, and ATFM methods are particularly prominent. Therefore, this paper focuses on flight delay propagation and discusses existing models and methods from two perspectives: delay propagation within airlines and delay propagation between airports, as shown in Figure 4.

3. Results and Discussion

3.1. Research on the Propagation of Flight Delays Within Airlines. The continuous increase of air traffic flow makes the problem of departure delay and arrival delay more and more serious [31, 32]. Departure delay and arrival delay are strongly correlated. Each flight in the system is repeated several times a day. Therefore, any interruption may delay the airline's subsequent flights [33], resulting in congestion in the airspace or other airports, queuing and delaying other airlines' flights [34]. Delays that happen earlier in the day often escalate significantly over time, meaning that delays on one flight can impact later flights, creating a domino effect that results in additional delays throughout the schedule [35]. Flights scheduled later in the day typically experience longer delays [36]. As shown in Figure 5 [37], after an aircraft experiences arrival delay A during its first flight, the delay will propagate to departure delay B at the departure airport, and the aircraft may experience arrival delay C at the arrival airport. Subsequently, downstream propagation may occur, forming a chain effect of flight delay propagation within the airline company.

Researchers have designed a series of methods to understand the mechanisms of delay propagation, identify factors affecting delay, and develop effective mitigation strategies. The research on the propagation mechanism of flight delays within airlines mainly focuses on how delays are spread through flight networks or flight chains [38]. This involves dependencies between flights (such as shared aircraft, crew, ground resources, and buffer time) and external disturbances (such as weather and airspace restrictions). The delay propagation prediction models mainly include machine learning models, data-driven models, and simulation models, which help alleviate the propagation of flight delays. The performance comparison table of the relevant research methods is shown in Table 2.



FIGURE 4: Flight delay propagation prediction method.

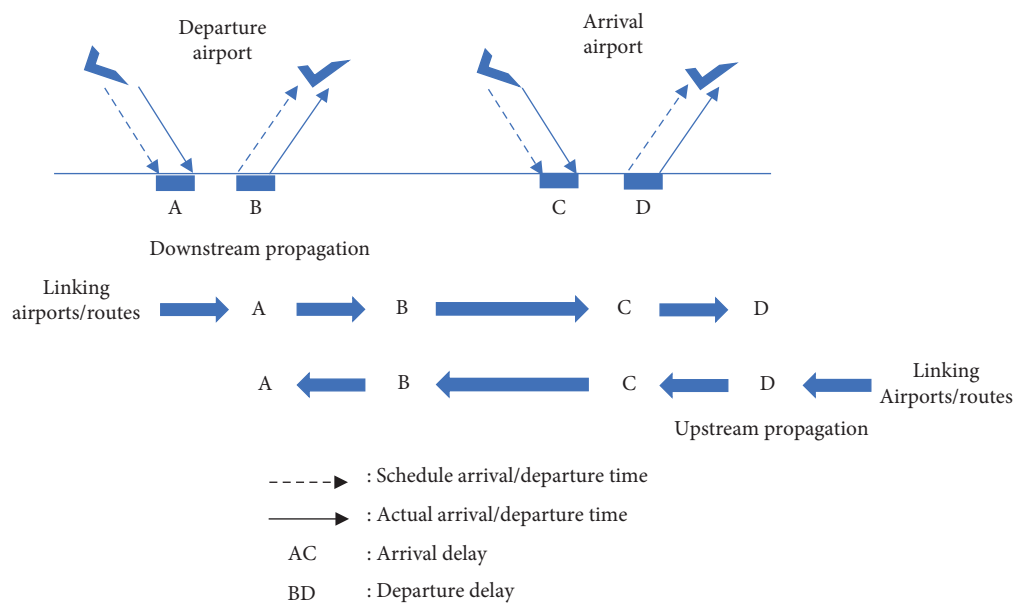
FIGURE 5: The process of delay propagation. The figure is copied from (<https://www.sciencedirect.com/science/article/pii/S0969699721000478?via%3Dihub>).

TABLE 2: Comparison table of performance indicators for delay models within airlines.

Model	Description	Application	Advantages	Limitations
Econometric model	Introduce relevant indicators to distinguish between propagation delay and newly generated delay	Analyze the transmission paths and influencing factors of delays, and optimize the flight schedule	Quantify the propagation of delays and newly generated delays, and optimize the buffer time	No further examination was conducted to reveal the differences in the propagation patterns of the departure and arrival nodes
Bayesian network model	Establish a probabilistic relationship to describe the impact of arrival delays on the propagation of flight delays	Analyze the impact of flight delays on subsequent flights	Provide an intuitive description of the scenario of delayed transmission and identify the weak links	Lack of consideration for turnover time and climatic factors
Function model	Determine the distribution of the delay sequence and test the propagation effect of the delay	Analyze the transmission effect of flight delays and optimize flight management	Effectively adjust the buffer time to make the flight schedule more reliable	It is impossible to determine the combined influence of multiple factors
Propagation trees	Analyze the potential for the spread of flight delays within the aviation network	Understand the dissemination of flight delays, and assess the depth and severity of the delay dissemination	Visualization and quantification of delay propagation	Failing to take into account the situation where multiple flight delays occur simultaneously
Delay prediction model based on machine learning methods	Predicting flight delays based on machine learning algorithms	Predict flight delays, identify key influencing factors, and optimize flight scheduling	Handle a large amount of data and achieve good prediction performance	The requirements for data quality and feature engineering are relatively high, and there may be a risk of overfitting
Real-time data-driven dynamic prediction method	A dynamic prediction model driven by real-time data, which updates the prediction results in real time	Real-time prediction of flight delays, adaptation to dynamic changes in flight status and environmental conditions	Real-time update of prediction results, strong adaptability, and improvement of timeliness and accuracy	The requirement for data update and processing speed is high. There may be problems related to data quality and noise
Simulation model	Simulate the dynamic process of flight delays	Evaluate the reliability of flight plans and design effective strategies for delay recovery	Simulate dynamic processes and provide intuitive decision support	The model construction is complex and the computing cost is high

3.1.1. Research on the Mechanism of Delay Propagation Within Airlines

3.1.1.1. Econometric Model. The use of buffer time by airlines to alleviate passengers' perception of flight delays is an effective way to mitigate the significant impact of flight delays. Without buffering, these delays would be caused by more complex network operations [39]. Scholars have used a new analytical econometric method to analyze the propagation patterns of downstream flight delays caused by upstream flight delays [40]. Firstly, an analytical model is used to quantify the propagation and newly formed delays that occur in each flight sequence flown by each aircraft within a day from three perspectives. Then, an econometric model is used and the Heckman two-step method is employed to reveal the impact of various influencing factors on the generation and progression of delay propagation. By analyzing the model, use equation (1) and (2) to identify the buffer amount in the flight and ground turnover plans, where equation (1) is the buffer zone on the flight link and equation (2) is the buffer amount in the ground turnover plan. Then, the propagation and newly formed delays are calculated from three perspectives, and it is found that there are differences in the delay propagation patterns among different airports and airlines, as shown in Figure 6. The research results indicate that departure nodes and arrival nodes have different delay propagation patterns. By effectively increasing buffer time, flight schedules can become more reliable and delay propagation can be reduced:

$$B_{i-1,i} = \max[0, (t_i^s - t_{i-1}^s) - U_{i-1,i}^F], \quad \forall i = 2, 4, \dots, I, \quad (1)$$

$$B_{i-1,i} = \max[0, (t_i^s - t_{i-1}^s) - U_{i-1,i}^G], \quad \forall i = 3, 5, \dots, I - 1. \quad (2)$$

To alleviate the propagation of delays, most airlines adjust their flights and ground buffer times. The selection of these buffer times is related to various factors, including the variability of flight times, the operating environment of flights, and the cost structure of airlines [41]. In addition, flight delays have temporal and spatial effects. Airlines will develop flight schedules several months in advance, and flight departure and arrival delays are intertwined with other flight-related times. As shown in Figure 7 [42], departure delay is the difference between the actual departure time and the planned departure time, and arrival delay is the same. Scholars have developed different econometric models to analyze flight departure and arrival delays, and have studied the impact of delay propagation on flight delays from the perspective of aircraft utilization, and investigated the effect of buffer time on flight delays [43]. In addition, the study also compares the propagation characteristics of flight delays in China during the departure and arrival phases, as well as the impact of operational, time, and weather factors on flight delays. It is found that departure and arrival delays are influenced by different factors, and airlines can more effectively reduce delays by inserting more air buffer time into their flight plans. Moreover, the utilization rate of aircraft has a significant impact on the propagation effect of delays,

especially when the flight sequence is later, the impact of early delays on flight punctuality performance is greater. Sensitivity analysis from the perspective of aircraft utilization shows that the impact of departure delay is greater than that of arrival delay.

In addition, some scholars have analyzed the main driving factors for Brazilian airlines to add extra time (i.e., "timetable buffer") to their flight schedules, evaluating the effectiveness of these extra times in improving flight on-time performance (OTP) [44]. Research has shown that an increase in extra time is associated with a decrease in the likelihood of flight delays, with an average reduction of approximately 13% for every minute of extra time added. However, research also shows that although the increase in extra time has a certain effect on improving flight punctuality, this schedule buffering behavior may mask the problem of low infrastructure efficiency, leading to sub-optimal allocation of resource utilization.

Similarly, for the delay of a single flight and its previous segment, considering the influence of factors such as operation, time, and weather, some scholars use econometric models to identify the propagation of flight delay and departure delay, explore the propagation characteristics of flight delay, and analyze the impact of departure delay on arrival delay [45]. Research has found that arrival delays are more likely to be affected by early delays and buffering effects compared to departure delays. When the arrival delay of incoming flights exceeds the departure buffer of subsequent flights, there will be a contagious departure delay. This study provides a quantitative method based on the delay propagation contribution of individual flights and their direct preceding flights. Unlike previous studies, this method does not attempt to trace the entire historical sequence of delay propagation, which has certain guiding significance for airlines to consider the delay propagation effect in flight scheduling decisions.

Econometric models mainly quantify delay propagation by introducing impact indicators related to flight delays, distinguishing between propagation delays and new delays. However, the economic model does not take into account the delay propagation caused by the connection between passengers and crew, as well as the connection resources, and does not further examine the differences in propagation patterns between departure and arrival nodes.

3.1.1.2. Bayesian Network Model. The flight delay propagation problem is a complex stochastic control system with multiple uncertain factors and their interactions. Bayesian methods can be used to study the interdependence between multiple factors in this complex system and establish probability relationships between node variables [46]. Establishing a Bayesian network model can intuitively describe the impact of arrival delays on flight delay propagation under different states.

Considering multiple connection sources such as aircraft, crew connections, and passenger connections within an airline, some scholars have developed a delay propagation model based on the Bayesian network for the airline network. As shown in Figure 8 [47], the model can identify

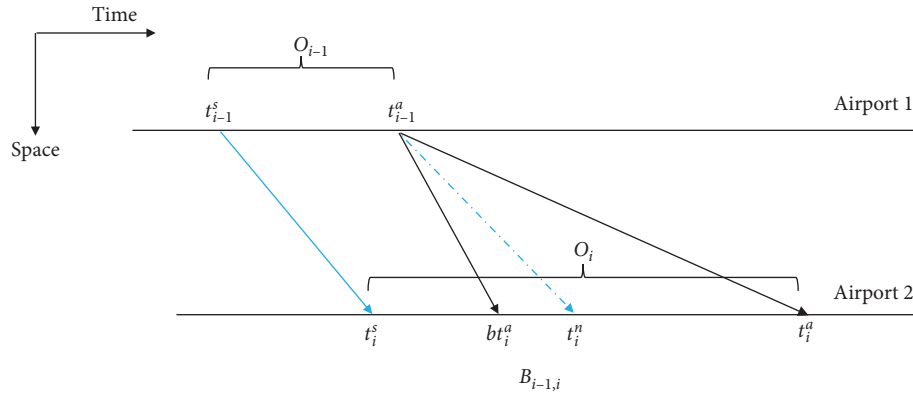


FIGURE 6: Time-space graph of a single flight. The figure is copied from (<https://www.sciencedirect.com/science/article/pii/S0191261515302010>).

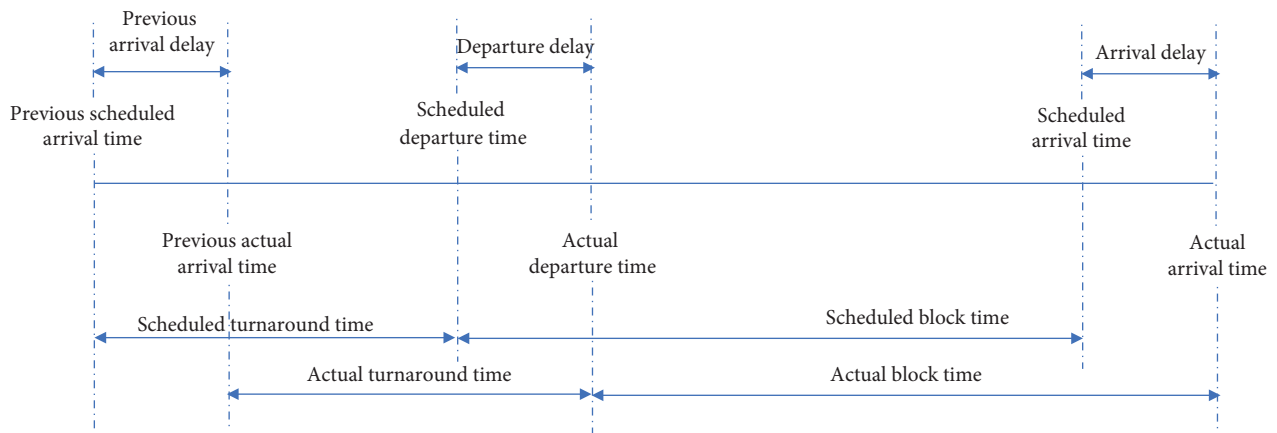


FIGURE 7: Time components related to a flight. The figure is copied from (https://scholarworks.sjsu.edu/cgi/viewcontent.cgi?article=3409&context=faculty_rsca).

weak links in the flight network within a delay tree framework and simulate delay propagation scenarios based on past operational data. Each node represents a flight, and each arc represents the relationship between two nodes, representing the aircraft, crew, or passengers. This Bayesian network model can overcome the limitations of modeling flight delay propagation in previous studies and provide a delay propagation model that is more in line with actual operational environments. By using two newly developed delay multiplier indicators (Expected Delay Multiplier, DME, and Expected Propagated Delay Multiplier, DMx), researchers can better capture the heterogeneous effects of flight delay propagation. The flight delay propagation model can clearly reflect the causes and propagation characteristics of flight delays, providing theoretical basis and practical guidance for airlines in flight scheduling and delay management.

The OTP of flights mainly depends on their arrival time. Therefore, it is crucial to use Bayesian networks to predict the arrival status of airports and examine the subsequent impact of these arrival delays on the next flight, especially for flights operated by the same airline. However, this method does not take into account the impact of turnaround time and climate factors on delayed propagation.

3.1.1.3. Function Model. To study the interdependence of flight delays under airline operations, weather, and air traffic control conditions and analyze the propagation effects of flight delays, some scholars have used Copula functions to determine the distribution of delay sequences and test the propagation effects of delays [48], as shown in equation (3). The tail dependence analysis of copulas is useful to measure the impact of upstream flight delays on downstream flight delays, as shown in equations (4) and (5), where λ^{up} and λ^{lo} represent the upper and lower tail dependencies (dependency coefficients) of the random variable, respectively. Research has found that flight delays can spread to downstream airports or airlines, where the intensity of the delay may weaken, spread, or increase. By effectively adjusting the buffer time, the flight schedule can be made more reliable. In addition, the study conducts sensitivity analysis to explore the impact of adding turnaround buffer time in flight planning on delay propagation and proposes specific buffer time adjustment suggestions for airlines and air traffic control factors. For flights that are frequently affected by airline factors, adding 20 min turnaround buffer time can reduce downstream delays. For flights that are frequently affected by air traffic control factors, adding 10 min of buffer time is sufficient to reduce delays. However, this may not

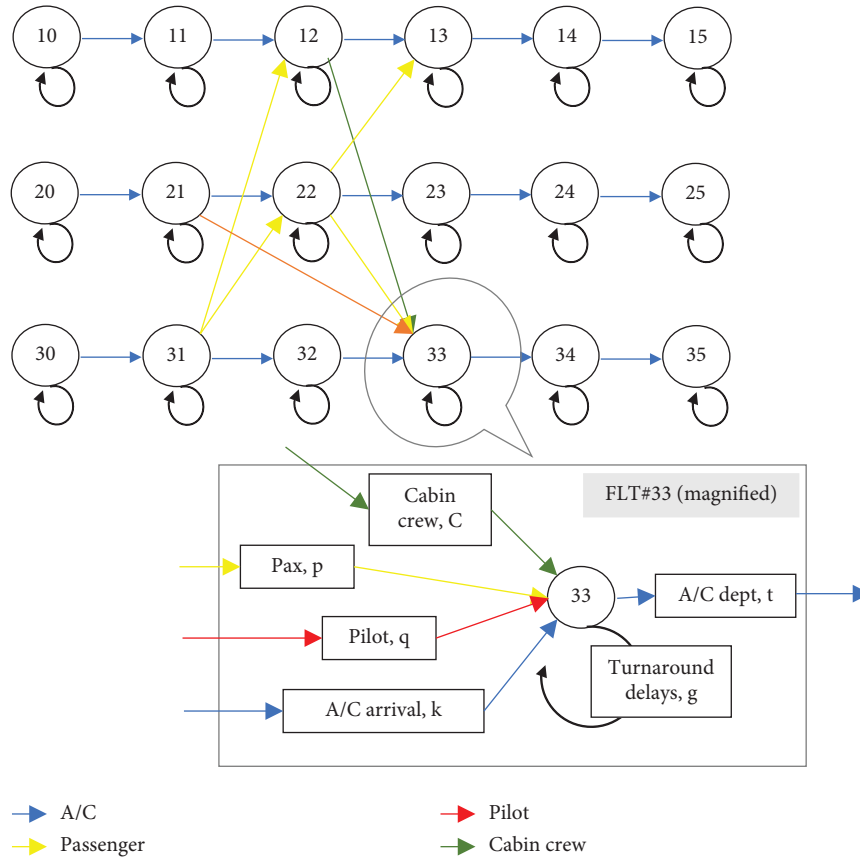


FIGURE 8: The causal relationship of delay sources in a DPT-BN model. The figure is copied from (<https://www.sciencedirect.com/science/article/pii/S1366554517309857#f0005>).

determine the possible joint impact of multiple factors. Therefore, it is necessary to further study the correlation of flight delays under comprehensive delay factors such as weather, airlines, and air traffic control in the future:

$$F(d_{i-1}^A, d_i^D) = C(F_1(d_{i-1}^A), F_2(d_i^D)), \quad (3)$$

$$\begin{aligned} \lambda^{\text{up}} &= \lim_{\alpha \rightarrow 1^-} \Pr[Y > F_2^{-1}(\alpha) | X > F_1^{-1}(\alpha)] \\ &= \lim_{\alpha \rightarrow 1^-} \frac{1 - 2\alpha + C(\alpha, \alpha)}{1 - \alpha}, \end{aligned} \quad (4)$$

$$\begin{aligned} \lambda^{\text{lo}} &= \lim_{\alpha \rightarrow 0^+} \Pr[Y \leq F_2^{-1}(\alpha) | X \leq F_1^{-1}(\alpha)] \\ &= \lim_{\alpha \rightarrow 0^+} \frac{C(\alpha, \alpha)}{\alpha}. \end{aligned} \quad (5)$$

Some scholars also use the probability density function to quantify the patterns of flight delays and propose two dynamic models to describe delay propagation: One is the model with a shifted power-law distribution, as shown in equation (6). Another type is the model with exponential truncated shifted power-law distribution, as shown in equation (7). Extract model parameters directly from data through data mining to characterize the operational efficiency of each airline in delay mitigation. This study reveals two common patterns of flight departure delay propagation.

By comparing the parameters of different airlines, their different performance in absorbing and mitigating delay propagation is observed [49]:

$$P(l_0 > l) = \frac{N(L)}{N_0} = c_0 (l + \beta)^{-\alpha_0}, \quad (6)$$

$$P(l_0 > l) = \frac{N(L)}{N_0} = c_2 e^{-(1/\lambda)} (l + \beta)^{-\gamma}. \quad (7)$$

Considering how specific flight delays interact with all downstream constraints in the corresponding aircraft rotation, some scholars have proposed a mathematically formulated cost function for flight-specific delays [50]. These functions have been reconstructed as random delay cost functions to consider the conditional probabilities and the increased uncertainty associated with further distance operation constraints. By learning conditional probabilities from historical operational data, the process of prioritizing flights can be supported to predict flight delay propagation patterns as part of tactical airline flight recovery. The research shows that the optimal recovery decision based on random delay cost allocates different flight schedules and reduces the number of passengers missing connections at hub airports. Therefore, decisions based on random models perform better in high delay scenarios than decisions based on deterministic delay costs.

The function model focuses on the propagation effects and response strategies of flight delays, mainly optimizing flight management from the aspects of adjusting buffer time, data mining, and probability evaluation. It explores delay patterns and mitigation measures, providing theoretical and practical references for understanding delay propagation mechanisms and optimizing flight management.

The introduction should be succinct, with no sub-headings. Limited figures may be included only if they are truly introductory, and contain no new results.

3.1.1.4. Propagation Trees. The propagation tree can analyze the propagation potential of flight delays in the aviation flight network and better understand the relationship between aircraft and crew scheduling and the operational performance of these schedules, especially how these schedules affect subsequent flights when flight delays occur. In addition, the propagation tree is a visual and quantitative tool used to understand how root delay propagates through the network without any other interference or scheduling modifications.

The propagation tree can analyze the propagation of flight delays, including the depth of delay propagation, the number of flights propagated, and the severity of delays, and propose some key indicators to quantify the impact of delay propagation. The influencing factors of flight delay propagation are highly complex and have nonlinear relationships [51]. In order to expand the complexity of the analysis, some scholars consider the probability of different root delays occurring, increase the propagation of crew and passenger connections, incorporate recovery decisions, and incorporate robustness indicators into the planning process. When an aircraft is delayed, it leads to delays in following flights until it is either taken for maintenance or enters an overnight period when no flights are planned. Likewise, the crew, whether cockpit or cabin, will cause delays for all following flights until they complete their duty period (or, if their overnight break is short, until their pairing ends). If the aircraft's original crew does not match the aircraft for the upcoming mission correctly, this may lead to a chain reaction in which other resources are also part of the delay propagation. As shown in Figure 9 [52], the aircraft designated for Flight 1 is rerouted to Flight 3. However, due to a 180-min initial delay, Flight 3 will experience a delay of 165 min. Once Flight 3 lands, the cockpit crew concludes their duty, and the aircraft proceeds to Flight 6. There is a 215-min gap between Flights 6 and 3, which means the delay will not affect Flight 6. Consequently, the 180-min initial delay from Flight 1 leads to an additional 430 min of delays for subsequent flights, resulting in four extra flight interruptions. Two different paths can be identified in this delay propagation diagram: $1 \rightarrow 2 \rightarrow 5 \rightarrow 7$ and $1 \rightarrow 3$. The longest path ($1 \rightarrow 2 \rightarrow 5 \rightarrow 7$) includes three additional flights beyond the initially delayed ones.

This approach considers the likelihood of various root delays happening, identifies the correlation between root delay groups, adds the propagation caused by the crew and connecting passengers, and incorporates recovery decisions, which can help strategically use the relaxation mechanism in

the system, thereby reducing the impact of outages, while enhancing the comprehension of delay propagation within the flight network. However, this method only considers aircraft and cockpit personnel, and other resources may cause additional delay propagation. In addition, the interaction between delays is not considered. There is usually a correlation between the delay and the propagation tree so that the propagation tree will affect each other. Moreover, the probability of root delay is not weighted, and the dataset is limited to two operators.

Due to the competition for runway resources between aircraft and crew waiting for early flights and delayed flights, downstream flights may experience delays. If there is not enough time to fully manage the interruption, the delayed connection will subsequently affect their outbound flights. The above model only focuses on one factor that causes flight delays to spread, which is a limitation of existing models. In order to reduce the propagation of operational delays without increasing planning costs and considering delays caused by shared resources such as aircraft, crew, and connecting passengers, some scholars propose a multilayer model for redistributing slack [53]. This model creates a propagation tree that explains all subsequent flights that may be affected by initial delays before rescheduling flights. By identifying the maximum delay of all incoming resources and adding it to the initial delay, the total propagation delay can be calculated.

The above model minimizes the total amount of propagation delay in the flight network by observing how each individual flight propagates delay. The main limitation of this method is that it does not take into account the fact that multiple flight delays may occur simultaneously in the network. In some cases, propagation delay may be inaccurately estimated.

3.1.2. Research on the Prediction of Delay Propagation Within Airlines

3.1.2.1. Delay Prediction Model Based on Machine Learning Methods. Flight delay propagation is a major social issue, and developing accurate delay propagation prediction models is an important component of mitigating flight delays [54]. In the research of flight delay propagation prediction, the application of machine learning is becoming increasingly common. Compared with traditional statistical methods, machine learning can process large amounts of data and make more informed decisions [55]. For example, some scholars propose a new flight delay prediction model based on machine learning that combines multilabel random forest classification and approximate delay propagation model. The hybrid model includes arrival delay prediction module, departure delay prediction module, and delay propagation module. The framework of the hybrid method is shown in Figure 10 [56]. The delay prediction model is the link between the arrival delay prediction module and the departure delay prediction module. Given the initial departure delay, establish a chain model by iteratively running the connection module. While analyzing the mode of delay

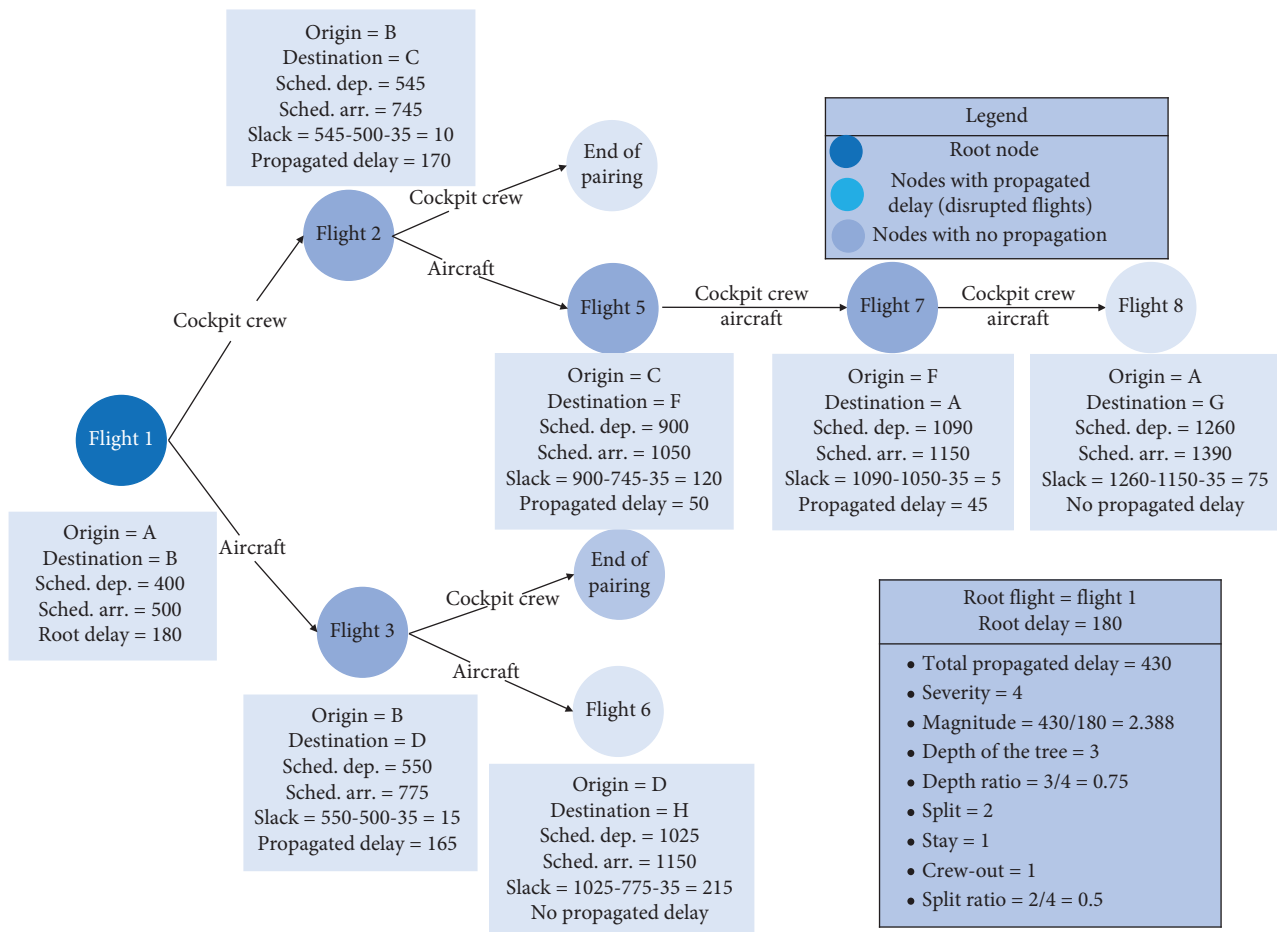


FIGURE 9: An example of a propagation tree. The figure is copied from (<https://web.mit.edu/sis07/www/cohn.pdf>).

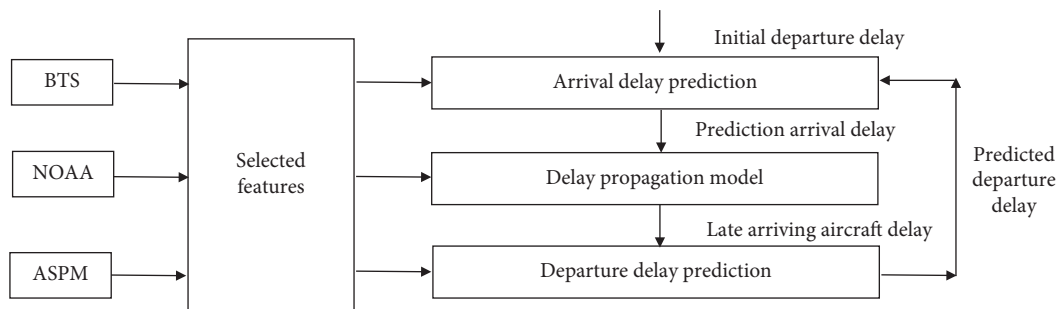


FIGURE 10: Mixed approach for chained delay prediction. The figure is copied from (https://www.researchgate.net/publication/330185077_Chained_Predictions_of_Flight_Delay_Using_Machine_Learning).

propagation, it is also possible to predict the delay status of flights along the same flight itinerary. This model introduces a feature selection process to improve the performance of multi label classification algorithms. The late arriving aircraft delay (LAAD) in the delay propagation model is used to describe the relationship between the previous arrival delay and the current departure delay. The chain model uses LAAD as a link to connect the previous arrival delay and the current departure delay, and iteratively predicts the delay of the same flight itinerary. Research has found that through the feature selection process, departure delay and late arrival

flight delay are the most important features in delay prediction. The chain model can predict flight delays along the same flight itinerary based on initial departure delays, and by updating actual departure delays, the accuracy of the model can be further improved. By developing predictive and normative analysis models, major flight delays can be predicted and flight schedules can be optimized one day in advance to reduce delays and improve the robustness of airline operations [57].

In addition, based on multifactor analysis, deep belief network (DBN) can uncover the inherent patterns of flight

delays. The combination of DBN and support vector regression (SVR) can address the challenges of processing large datasets and capture key factors affecting delays. Therefore, some scholars have used DBN methods to explore the intrinsic patterns of flight delays and embedded SVR into the model for supervised fine-tuning to improve the accuracy of predictions. The structural framework of the DBN-SVR model is shown in Figure 11 [58]. DBN is used to extract the main factors affecting flight delays, and then, the output of DBN is used as the input of the SVR model to capture the key influencing factors causing flight delays and generate delay prediction values. In the analysis of key influencing factors, it is found that the delay before the flight, route conditions, and airport congestion are the key factors affecting flight delays. The DBN-SVR model proposed in this study can effectively predict flight delays and can be embedded into airport flight information systems, integrated with existing delay prediction engines, to help airports and airlines improve service quality, reduce passenger anxiety and complaints. However, the model also has certain limitations, with relatively single data, no use of domestic and foreign flight information to improve the accuracy of the model, and no special research on the deep patterns of cargo flight delays.

The flight delay prediction method based on deep learning can establish a chain model of flight delay propagation by analyzing the chain propagation characteristics of flight delay. Therefore, some scholars have proposed two improved models, CBAM-CondenseNet network and SimAM-CNN-MLSTM network, for predicting flight delays [59]. CondenseNet is a densely connected network based on convolutional neural networks, which is improved by inserting CBAM modules into the network. Extract features from the fused flight chain data to make it more suitable for the task of predicting flight delay propagation. The SimAM-CNN-MLSTM network not only considers the spatial characteristics of flight missions but also focuses on the temporal relationships between flight chain data. The model also uses an attention mechanism module to enhance important neurons in the feature matrix. The improved model has significant advantages in handling flight delays and prediction tasks. Through analysis, it is found that the two proposed deep learning models can effectively predict flight delay propagation, among which SimAM-CNN-MLSTM network performs the best in accuracy prediction.

In the study of flight delay propagation, the flight delay prediction method based on machine learning mainly refers to the prediction of flight delays and the identification of key influencing factors affecting flight delays. In addition, machine learning algorithms can also estimate the probability distribution of individual flight delays, which can support planners to plan flight operations robustly [60].

3.1.2.2. Real-Time Data-Driven Dynamic Prediction Method.

Flight delay propagation is dynamic, and delay data, processes, and models will change over time and be updated in real time. The delay propagation process has temporal and nonlinear characteristics, covering the collection and processing of time-varying data such as real-time flight status,

weather conditions, and airport operation status. At the same time, the model is required to adapt to these constantly changing inputs and update prediction results in real time. This dynamism not only reflects the immediacy and complexity of delay propagation but also makes research more closely related to actual aviation operation scenarios, providing support for real-time decision-making. Predicting real-time flight delays involves continuously estimating the current state of the system [61]. Thanks to the advancement of big data storage technology, the civil aviation department has collected a large number of flights and related weather data [62]. This information can significantly enhance the precision and effectiveness of predicting delay propagation. A flight delay prediction framework utilizing a Kalman filter-based dynamic data-driven approach can analyze the delay propagation between successive arriving flights. As shown in Figure 12 [63], various inputs, including flight data, arrival schedules, airport status, and weather conditions, are collected and organized. An online parameter estimation module assesses the cumulative delay, buffer time, and real-time measurement data to enhance the adaptability of the state space model. The system state module computes the preliminary landing delay estimates for upcoming flights, while the data assimilation module refines these estimates using real-time data. Experiments on historical flight data show that the model achieves high prediction accuracy, which remains largely unaffected by the number of consecutive flights.

By constructing a state space model to analyze the composition of the landing delay, it is assumed that all delays before the flight enters the landing preparation stage are cumulative delays, and the delays generated once the flight enters the preparation stage are classified explicitly as landing delays. The sum of these two types of delays together constitutes the total arrival delay of the flight. To quantify this process, two basic equations are used: Equation (8) is used as the process equation, which describes how the cumulative delay and landing delay evolve and various factors (such as buffer time and noise) from the $k-1$ th flight to the k th flight; equation (9) is used as the measurement equation, which reflects how to estimate or correct these delay states through real-time observation data v_k , where x_k represents the landing delay of the k th flight, c_k represents the total cumulative delay accumulated to the flight, b_k represents the buffer time between two consecutive flights, and w_k and v_k represent the uncertainty or noise introduced during the process and measurement, respectively:

$$x_k = \begin{cases} x_{k-1} + c_{k-1} - b_k + m + \varepsilon_k + w_k, & x_{k-1} + c_{k-1} \geq b_k - m, \\ 0 + \varepsilon_k + w_k, & x_{k-1} + c_{k-1} < b_k - m, \end{cases} \quad (8)$$

$$y_k = x_k + v_k. \quad (9)$$

This method can be further studied in three aspects: improving the state space model to use additional real-time data, using existing or measurable data to estimate unmeasurable variables, and adapting noise variance.

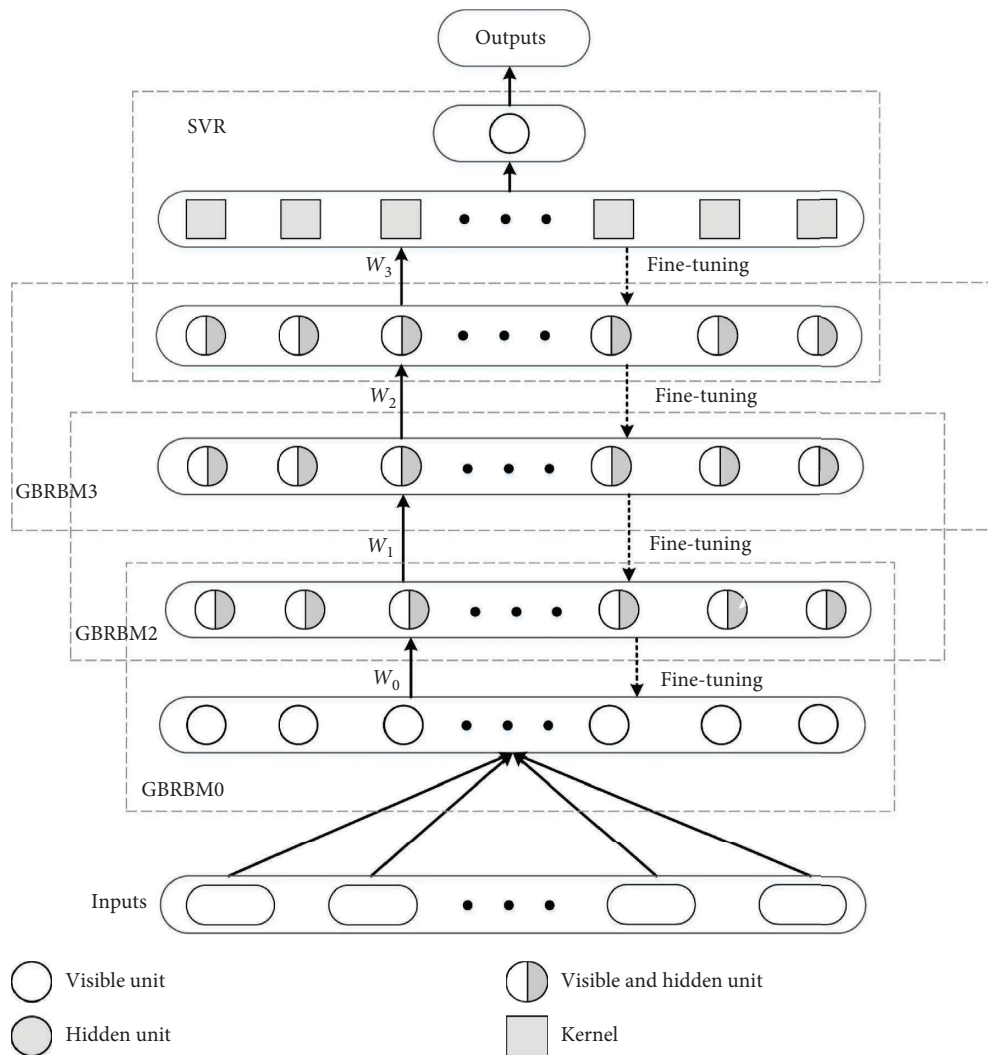


FIGURE 11: Structure of the employ DBN-SVR model. The figure is copied from (<https://www.sciencedirect.com/science/article/pii/S1366554518311979>).

3.1.2.3. Simulation Model. Flight delay propagation is a complex dynamic phenomenon, which not only affects the reliability of flight plans but also has a profound impact on the operating costs of airlines, passenger satisfaction, and overall efficiency of air traffic system. Simulation models, as a flexible and effective analysis tool, are widely used to study the flight delay propagation mechanism and its solution design [64, 65]. The simulation model can predict delay propagation by simulating flight delays [66–68] and analyze the impact of factors such as flight scheduling, flight connectivity, and airport congestion on delay propagation [69].

Researchers have developed a model based on agents, which includes a conditional probability model for modifying the expected departure time and determining whether a flight has undergone the necessary waiting time due to ground delay plans (GDP) or airline-related reasons [70]. In the agent-based delay prediction model, two random forest regression models are introduced to estimate the turnaround time-of-flight agents and the passage time-of-flight agents, as shown in Figure 13. This model can be used to predict the delay of a single flight in the tactical ATFM stage by directly

incorporating real-time information such as flight plans, flight status, and weather conditions into the model. The study shows that the performance of this model is superior to existing research, and further explores the positive impact of introducing parameter models on the prediction performance and the negative impact of increasing the prediction range on the prediction performance.

Simulation models are widely used in the study of flight delay propagation to evaluate the reliability of flight plans and design effective delay recovery strategies. Through methods such as multiagent simulation, statistical analysis, and mixed simulation, researchers can quantify the impact of delay propagation, optimize buffer time allocation, and propose targeted solutions. In addition, simulation models also provide important support for airline operational performance evaluation and air traffic management decision-making. In the future, by combining multisource data fusion, artificial intelligence, and real-time dynamic simulation technology, the simulation model will further enhance the accuracy and practical application value of flight delay propagation research.

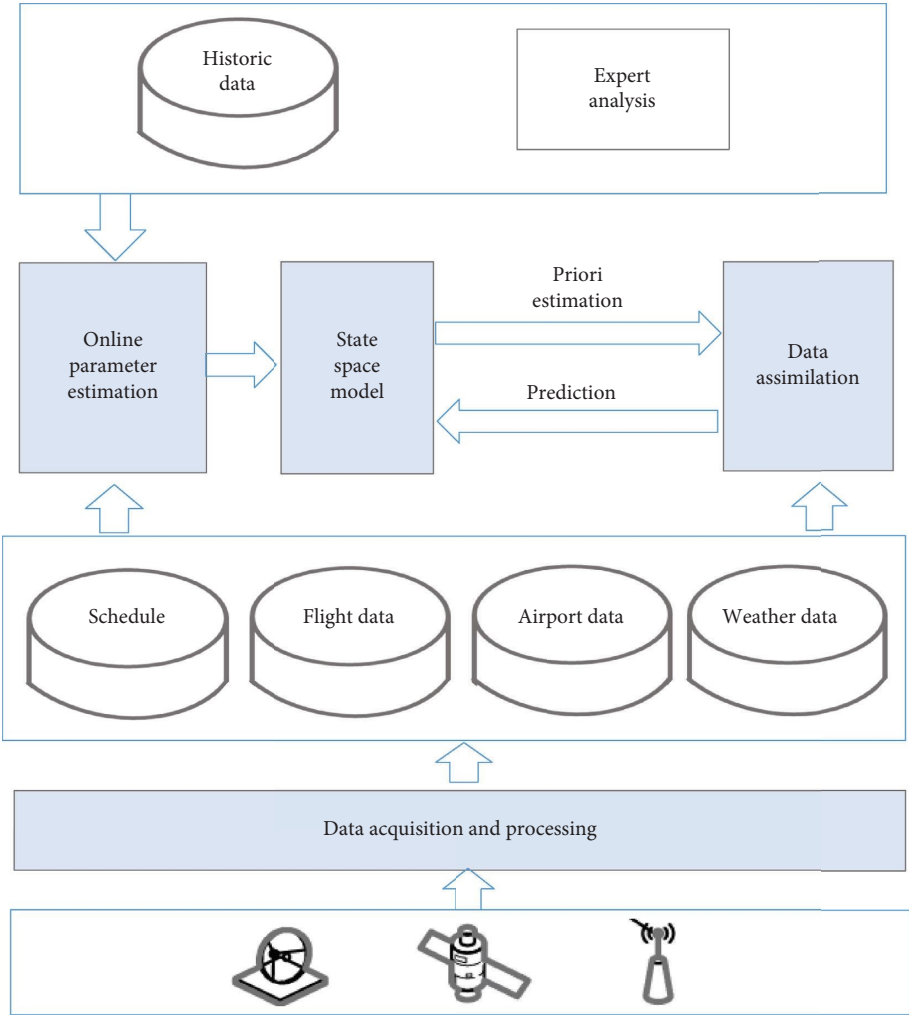


FIGURE 12: Dynamic data-driven application system architecture for flight delay. The figure is copied from (<https://www.jsoftware.us/vol7/jsw0702-03.pdf>).

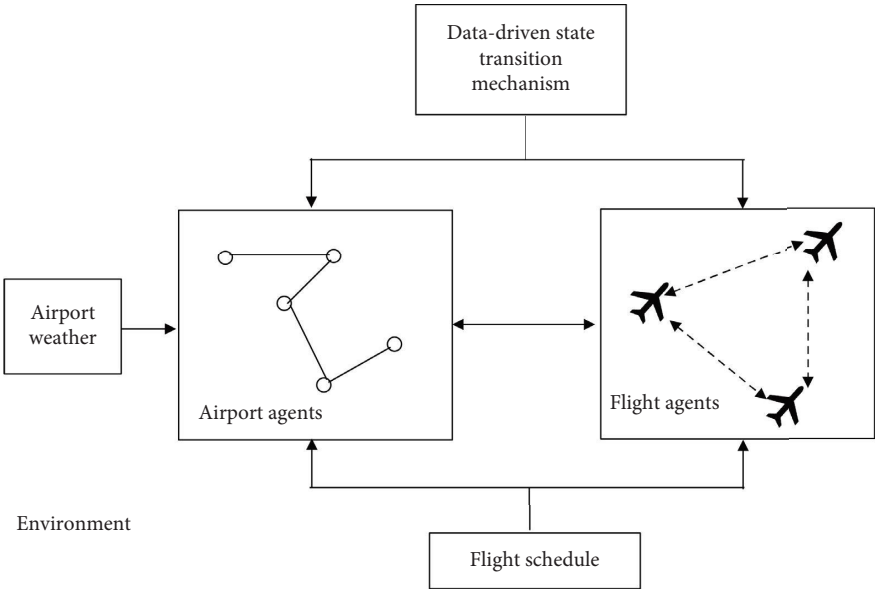


FIGURE 13: The architecture of the agent-based flight delay prediction model. The figure is copied from (<https://pdfs.semanticscholar.org/5588/c9d437f845e49cebd3a69b63ba3910cced7b.pdf>).

3.2. Research on the Propagation of Flight Delays Between Airports. The air transport network, also known as the airport network, is a complex system that includes numerous airports, each serving different functions within the air transport framework. An airport's specific conditions, such as flight delays or congestion [71], can significantly influence the overall network. If there is not enough buffer time to accommodate an initial delay, it can propagate throughout the network [72]. This means that subsequent flights may experience delays while waiting for the aircraft and crew from the delayed flight, a phenomenon referred to as delay propagation. In many cases, airlines can speed up ground operations to help delayed flights return to normal. It is also expected to establish some idle time in the running plan to adapt to typical changes in block time and absorb moderate running delays. However, for example, if the first leg of a multileg flight is delayed by 1 hour, then the second and third legs of the flight are likely to remain behind schedule, assuming that they are not canceled, even if there are no subsequent operational delays, delays in the following leg will still occur [73, 74].

In reality, flight delays tend to spread horizontally between airports more often than vertically within a single airport. The entire flight network exhibits numerous hidden and dynamic heterogeneous correlations between airports, making it more challenging to predict than the other two levels. Many flights arrive at an airport, utilizing its support, apron, and runway resources, and they must wait for the planes to take off and land in a specific sequence [75]. The scarcity of parking spots and runways can cause disruptions in an airport's schedule when delayed flights arrive or depart, resulting in a mismatch between the supply and demand for different flight resources, such as airport capacity and air traffic. As a result, other flights may also experience delays, highlighting how flight delays can propagate across airports. To mitigate the propagation of these delays, it is essential to conduct a detailed analysis of the airport network's structural features and explore the intricate connections between flight delays and various transportation resources. To better understand how flight delays are transmitted at the system level—specifically, how one airport impacts others—researchers have started to develop delay propagation networks and examine their characteristics from a network standpoint. The performance comparison table of the relevant research methods is shown in Table 3.

3.2.1. Research on the Mechanism of Delay Propagation Between Airports

3.2.1.1. Time Interval Petri Net (TIPN). In order to understand the changing nature of flight delay propagation, some scholars use the TIPN to represent flight operation and check how delays are propagated within the aviation sector. Petri nets are commonly utilized in modeling discrete event dynamic systems, as well as for performance assessment, scheduling, and control. The model combines the time interval constraint to represent the flight turnaround time and flight cycle and considers the different initial delay levels of

the source airport, which can effectively predict the flight operation status and delay level of the downstream airport in real-time. The analysis of the aircraft operation process, illustrated in Figure 14 [76], reveals that a small initial delay time only impacts a few downstream airports. As the initial delay time increases, more downstream airports are affected. However, once the initial delay time surpasses a certain threshold, it begins to influence all downstream airports, leading to a reduction in the delay time experienced by each airport. Given that flight operations are significantly influenced by unpredictable conditions, it is essential to enhance the feasibility and real-time capabilities of the flight delay propagation analysis algorithm.

Airport traffic has an essential impact on delay propagation. In particular, airports with large traffic volumes are more prone to delays, which will quickly spread to other airports. Airports that experience high levels of traffic significantly contribute to the delay in the propagation, because delays can easily spread from these airports and can quickly spread to most airports. Therefore, before the delay begins to spread from these airports, it is necessary to control airports with large airport traffic effectively. Furthermore, as the network structure evolves and more tier and regional airports are established, the primary effect of delay propagation is increasingly moving from hub airports to these secondary and regional airports. The increase in these airports does not match resources such as runways and terminals. Consequently, when these airports become congested, it takes longer to recover. Airports play a vital role in managing air traffic flow, but they have also become a limiting factor for both air traffic management and safety operations at the airport [77].

3.2.1.2. Cox Proportional Hazards Model. To examine how delay factors affect flight delays, the Cox proportional hazards model is used to simulate delays in multi-airport environments [78]:

$$h(t|X) = h_0(t, \alpha) \exp(\beta'X). \quad (10)$$

Equation (10) not only constructs a mathematical model for comprehensively assessing the risk of flight delays but also identifies several key factors (i.e., explanatory variables) that work together in predicting flight delays. At the same time, equation (11) is particularly cleverly designed. It allows direct quantitative analysis of how the percentage of flight delay risk will change when a specific variable x_k increases by one unit under the premise of controlling all other variables. This change may be manifested as an increase or decrease in risk, thus providing a unique perspective for an in-depth understanding of flight delay risk and its driving factors:

$$\frac{\hat{h}(t|x_k + 1) - \hat{h}(t|x_k)}{\hat{h}(t|x_k)} = e^{\hat{\beta}_k - 1}. \quad (11)$$

This model differs from earlier simulation models and statistical analysis techniques as it can capture the dynamic nature of flight delays. Cox regression analysis identifies the main factors contributing to take-off and arrival delays. The

TABLE 3: Comparison table of performance indicators for delay models between airports.

Model	Description	Application	Advantages	Limitations
Time interval Petri net	Represent flight operations using time interval Petri nets and analyze the propagation of delays	Real-time prediction of the flight operation status and delay levels of the downstream airports	Predict flight operation status and delay levels, with strong adaptability	It is necessary to enhance the feasibility and real-time performance of the algorithm
Cox proportional hazards model	Simulate delays in multiairport environments and identify key factors	Assess the risk of flight delays and optimize resource allocation	Capture the dynamic characteristics of flight delays	The data processing procedure is complex
Queuing network model	The propagation of simulated flight delays in multiairport systems	Analyze the impact of airport congestion on flight delays and optimize flight scheduling	Typical queue length and waiting time for accurately predicting flight delays, with strong adaptability	High requirements for data update and processing speed
Model based on complex network theory	Analyze the propagation paths and mechanisms of delays among airports	Evaluate the structural characteristics of the airport network, and optimize flight scheduling and resource allocation	Identify key airports and the transmission paths of delays	No further research has been conducted on the mechanism of delay propagation at individual airports or for specific flights
Real-time data-driven dynamic prediction method	A dynamic prediction model driven by real-time data, which updates the prediction results in real time	Real-time prediction of flight delays, adaptation to dynamic changes in flight status and environmental conditions	Real-time update of prediction results, highly adaptable	High requirements for data update and processing speed
Delay propagation prediction model based on machine learning	Predicting flight delays based on machine learning algorithms	Predict flight delays, identify key influencing factors, and optimize flight scheduling	Handle a large amount of data and achieve good prediction performance	The requirements for data quality and feature engineering are quite high
LSTM model based on deep learning	Predicting flight delays using deep learning algorithms	Predict flight delays, handle high-dimensional feature data, and enhance prediction accuracy	Extract complex feature patterns to enhance prediction accuracy	The model has a high complexity and requires a long training time
Spatiotemporal graph neural networks	Utilizing spatiotemporal graph neural networks and integrating spatiotemporal data, predict flight delays	Predict flight delays, handle complex spatiotemporal data, and enhance prediction accuracy	Effectively integrate spatial correlation and temporal dynamics, and provide accurate predictions	High requirements for data quality and network structure
Air traffic flow management	Optimize flight scheduling by using different algorithms or multiobjective optimization methods	Optimize flight scheduling to reduce delays and conflicts	Provide the optimal solution and significantly enhance performance	The cost of calculation is high and the model construction is difficult

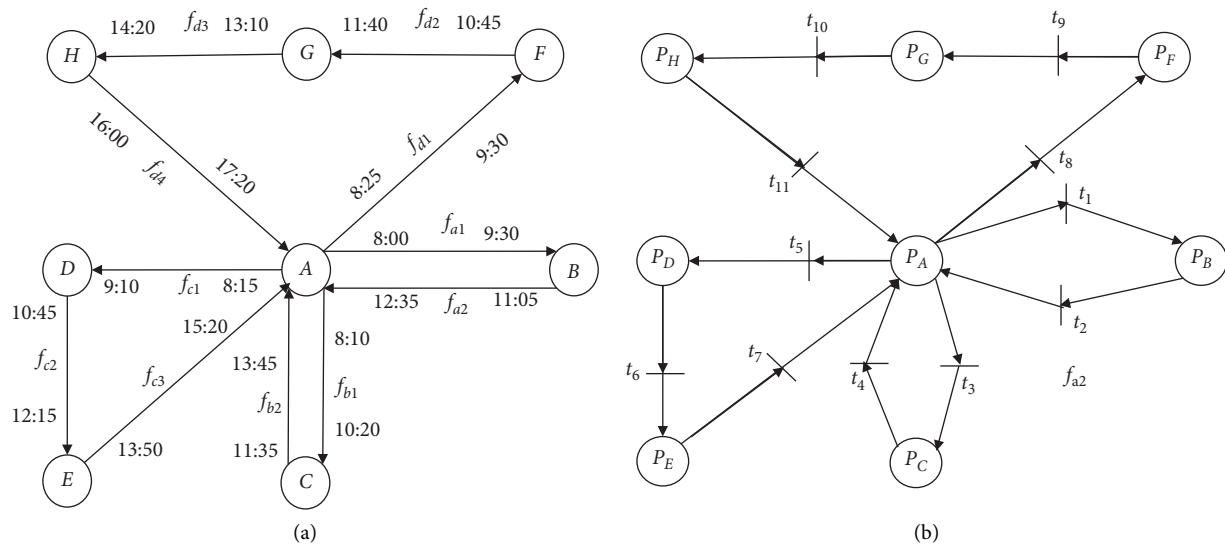


FIGURE 14: Flight information and its TIPN model. (a) Flight information graph for aircraft. (b) TIPN model for operation. The figure is copied from (<https://benthamopen.com/contents/pdf/TOAUTOCJ/TOAUTOCJ-6-433.pdf>).

resulting risk ratio enables airlines to evaluate their recovery abilities and make informed decisions about resource allocation to ensure efficient scheduling.

3.2.1.3. Queuing Network Model. Decisions about airline scheduling and resource allocation have a significant impact on airport congestion, which affects the overall travel time. From a spatial viewpoint, the level of airport congestion (the volume of flights at the airport) and the configuration of the aviation network (the network's topological characteristics) are the primary elements that lead to flight delays [79]. Since the flight networks of most airlines are closely interconnected and resources like airports are utilized by multiple networks, any delays that occur within the system—whether from individual flight delays or ground stops at the airport—can influence the entire transportation network, impacting the overall efficiency of the air transport system [80, 81].

To examine the propagation of delays in air traffic networks, a data-driven model is constructed to simulate the actual observed delay development trend and covers three core mechanisms of delay propagation: aircraft rotation scheduling, passenger or crew flight connection, and airport congestion. Studying how factors such as passenger connectivity and airport capacity affect the delay propagation of the system can quantify the degree of delay propagation in the network [69]. In addition, using the queuing network model, the propagation of flight delays within multiairport systems can also be analyzed by point-by-point steady-state queuing, by calculating the unimpeded movement time between different servers to estimate the propagation of flight delays, and then accurately predict the typical queue length and waiting time associated with flight delays, mirroring the accurate operation [82]. Moreover, studying current traffic data can create a data-driven queuing network model focused on airports [83]. The model is utilized to

simulate how the delay propagates throughout the network, notably when airport capacity is diminished due to local disruptions like weather conditions or air traffic controller strikes. If the airport's capacity is lower than the airport's specific critical capacity value, the total network delay will increase significantly.

3.2.1.4. Model Based on Complex Network Theory. There are many airports in the air transport system, and their interactions are complex. Information from a single airport level alone cannot provide a deeper understanding of delay propagation. Delay propagation occurs due to processes that operate across various time scales [84]. Analyzing complex networks can assist in reducing the propagation of delays by pinpointing the routes through which delays travel between airports [85].

Dynamic complex network (DCN) analyzes the delay interactions among various airports by considering the interrelated delay time series. It illustrates the way and extent of delay propagation and helps to understand the overall structure and dynamic characteristics of delay propagation within the airport system. In this framework, edges signify daily time intervals, indicating the functional connectivity and possible operational conditions between airports. By developing a DCN, air traffic managers and planners can enhance decision-making, boost the efficiency of aviation systems, and minimize delays.

Figure 15 gives a network example to illustrate the network analysis method [86]. In Figure 15(a), the example network consists of 8 airports ($N = 8$) and 12 edges ($M = 12$). The average degree of the network is 1.5, which means that each airport has an average impact on another 1.5 airports. The degree of an airport reflects how many airports it has a delayed propagation link with. For example, in Figure 15(b), Airport 1 is influenced by Airport 3 and, in turn, influences both Airport 2 and Airport 3. Figure 15(c) shows

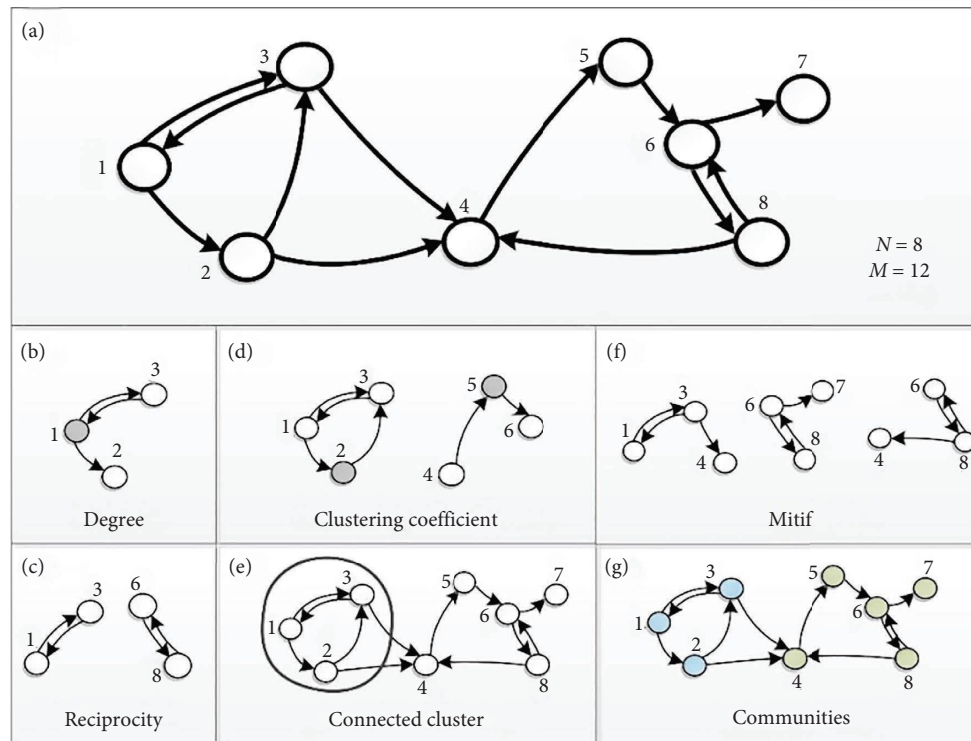


FIGURE 15: Features of a sample network. The figure is copied from (<https://www.sciencedirect.com/science/article/pii/S1366554518301042#f0010>).

the nature of reciprocity, that is, Airport i has an effect on Airport j , and Airport j will have an impact on Airport i . The parameter R can be used to describe the overall symmetry of the directed network quantitatively, and its maximum value is 1, indicating that the delay propagation between all airports is interoperability. The larger the R , the greater the symmetry of the network. The agglomeration coefficient of the airport measures the proportion of directly connected links with delayed propagation and the number of triangles in the network between adjacent airports, as shown in Figure 15(d). The two adjacent airports in Airport 2 interact with each other to form a small group, while the two adjacent airports in Airport 5 do not establish a connection. Therefore, the clustering coefficient of Node 2 is greater than that of Node 5. The largest connected cluster is composed of airports connected by transmission, as shown in Figure 15(e). The most significant connected clusters include Airports 1, 2, and 3. The community is utilized to assess whether the delay propagation between airports can be segmented into multiple subregions. Each subregion has the characteristics of the high density of delay propagation links in the airport and less contact with the large-scale system. Figure 15(g) illustrates two cells of the sample network. In addition, the purpose of modularization is to quantify the robustness of the network when it is divided into the above cells. Figure 15(f) demonstrates that the relationship patterns among the three groups of airports are alike.

The connection between the in-degree and out-degree of airports within the DCN network indicates that several significant airports play a role in reducing the delay

propagation path. The reciprocity parameter indicates that delay propagation among DCN airports occurs in both directions. In addition, community and modularity indicators help to determine whether delayed propagation can be divided into multiple subregions and identify clusters formed by major airports. Furthermore, there is a significant correlation between the average daily flight delays and the maximum connected cluster. This cluster of airports will evolve considerably over time, which indicates that the origin of the delay propagation does not come from the same group of airports.

This approach assesses the time series of arrival delays at various airports, examines the global relationships of delay propagation, and develops and analyzes the mechanisms behind flight delays. It utilizes complex network theory and relevant metrics to offer theoretical backing for creating strategies to reduce delay propagation. However, in investigating the relationships of flight delay propagation, this method primarily emphasizes the overall aviation network perspective. It does not further study individual airports or specific flights, potentially resulting in a limited understanding of local delay propagation mechanisms [87].

In addition, the extreme delay propagation network (EDPN) can use the principle and measurement of complex network theory to test the degree, classification, network efficiency, community structure, and network mode of delay propagation path; analyze the characteristics of delay propagation path; and determine the flight segments that need attention, so as to give priority to high-frequency flights in air traffic management to slow down the propagation of

delay. In addition, the network involves a new airport identification index, which uses complex network theory to study the features of EDPN. This provides theoretical support for air traffic management officials to develop effective strategies, allocate resources wisely, and improve the resilience and operational effectiveness of the overall system [88].

3.2.2. Research on the Prediction of Delay Propagation Between Airports

3.2.2.1. Real-Time Data-Driven Dynamic Prediction Method. With the development of real-time data acquisition technology, more and more researches have begun to focus on using real-time data to drive dynamic prediction [89]. For example, obtaining real-time weather data, airport operation data, and other information can dynamically update the parameters and variables in the prediction model and realize real-time prediction and dynamic adjustment of flight delay propagation. This method can better adapt to the uncertainty factors in actual operation and improve the accuracy and timeliness of prediction. However, unlike the internal prediction methods of airlines, it requires more efficient data processing and analysis capabilities, which puts forward higher requirements for technical implementation and system performance.

Therefore, an analysis model known as approximate network delay (AND) is proposed, which focuses on queue and network decomposition. The model examines the propagation mode of delays on the vast network of significant airports, evaluates the delays caused by local congestion at a single airport, and identifies the chain reactions that cause delays to spread. The model alternately uses the queuing engine (QE) and the delay propagation algorithm (DPA) to determine the delay of each airport. As shown in Figure 16 [90], DPA adjusts the demand profile of each aircraft on the delayed route of the upstream airport by monitoring the impact of local delays on a single aircraft and realizes the propagation of delays from one airport to other areas of the network. Meanwhile, the QE operates at each airport within the network to estimate the anticipated delays for each flight's take-off and landing. The DPA assesses whether a flight delay is significant enough to cause a ripple effect, affecting the initial demand rate for any specific time frame at any airport in the network. The model offers valuable insights into how delays spread throughout the network, influencing flight schedules and demand rates.

The AND model does not take into account delays caused by route congestion, airlines' response to congestion, and additional causes of delays, like issues with aircraft mechanics, in solving delay propagation. Additionally, the main criterion for evaluating the AND model is the model's ability to replicate observed trends and behaviors in the NAS system rather than its ability to match the actual measured delays perfectly.

Therefore, to illustrate the connection between delay propagation and network structure, highlight how airport connectivity and traffic affect flight delay propagation across

various air traffic networks [91, 92]. A data-driven air transport delay propagation model based on the epidemic process model is proposed [93]. This approach involves creating two data-driven epidemic models: One focused on flights and the other on airports. By drawing a parallel line between the propagation of disease and the propagation of delay, the simulation of the propagation of flight delays in the network is realized. In addition, the cross-log model is used to determine the probability of delay propagation [94]. For example, the FSIS model uses the epidemic transmission mechanism found in the SIS model to study how flight delays are transmitted. In this model, the rules governing the propagation of flight delays and the actual probability of delays are redefined to study how delays propagate within the air transport network. It can evaluate various factors that influence the likelihood of delay propagation, including flight frequency, route distance, scheduled buffer time, and duration of propagation delay.

The prediction of airport delay propagation based on the SIRS model regards the delay propagation in the airport network as the propagation process of an infectious disease, including three individuals: susceptible (S), infected (I), and recovered (R). The susceptible person will be infected when in contact with the source of infection. The infected person can be transmitted to the vulnerable person. The infected person will recover with the cure rate and have immunity. In some cases, the recovered person will lose immunity and become susceptible again. The infection mechanism is shown in Figure 17 [95]. The model simulates the propagation process of airport delays by describing the infection mechanism. In the airport network, delays can occur due to limited capacity, equipment malfunctions, and severe weather conditions. When delays affect flights that share resources, they can spread from the departing airport's upstream flight to the arrival airport. This delay propagation is represented in a model as a graph, where vertices symbolize airports and edges indicate their connections. If one airport experiences significant delays, it can lead to increased delays at connected airports. Furthermore, while delays can be mitigated in later operations and are not influenced by delays at the same starting airport, they can still be impacted by delays at different initial airports. Considering the complexity of the airport network, the evolution of delays exhibits typical complex network propagation characteristics, as illustrated in Figure 18.

Compared with the uniform SIS propagation model, the heterogeneous SIS propagation process is used to simulate the propagation of airport congestion, which is realized by introducing different propagation characteristics on the aviation network. The heterogeneous SIS model is proposed better to capture the complexity of congestion propagation between airports and more accurately assess the vulnerability of airports. The heterogeneous SIS model studies airport congestion through various SIS propagation processes based on airline networks, defines airport vulnerability as the likelihood of congestion at each airport, and creates three distinct airline networks to reflect various flight characteristics. In this model, the infection rate and recovery rate vary with the network structure of the airport. The

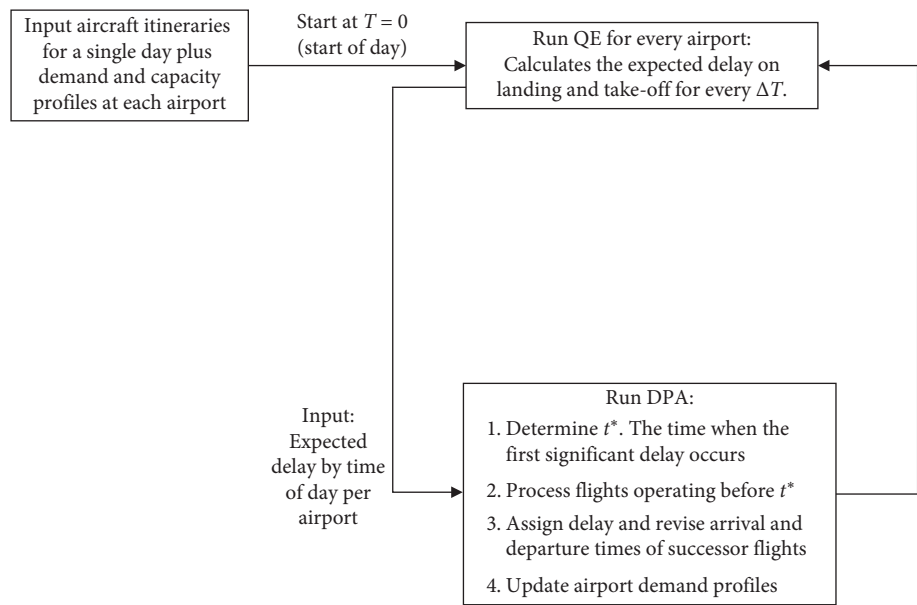


FIGURE 16: A schematic representation of the operation of the AND model. The figure is copied from (<https://www.sciencedirect.com/science/article/pii/S0968090X11000878#f0005>).

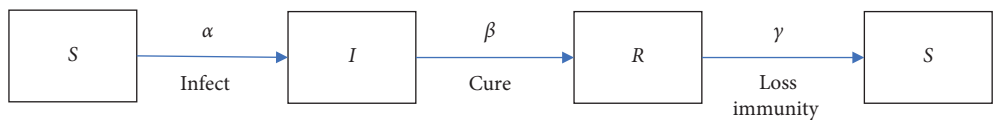


FIGURE 17: Infectious mechanism of SIRS. The figure is copied from (<https://onlinelibrary.wiley.com/doi/epdf/10.1155/2020/8816615>).

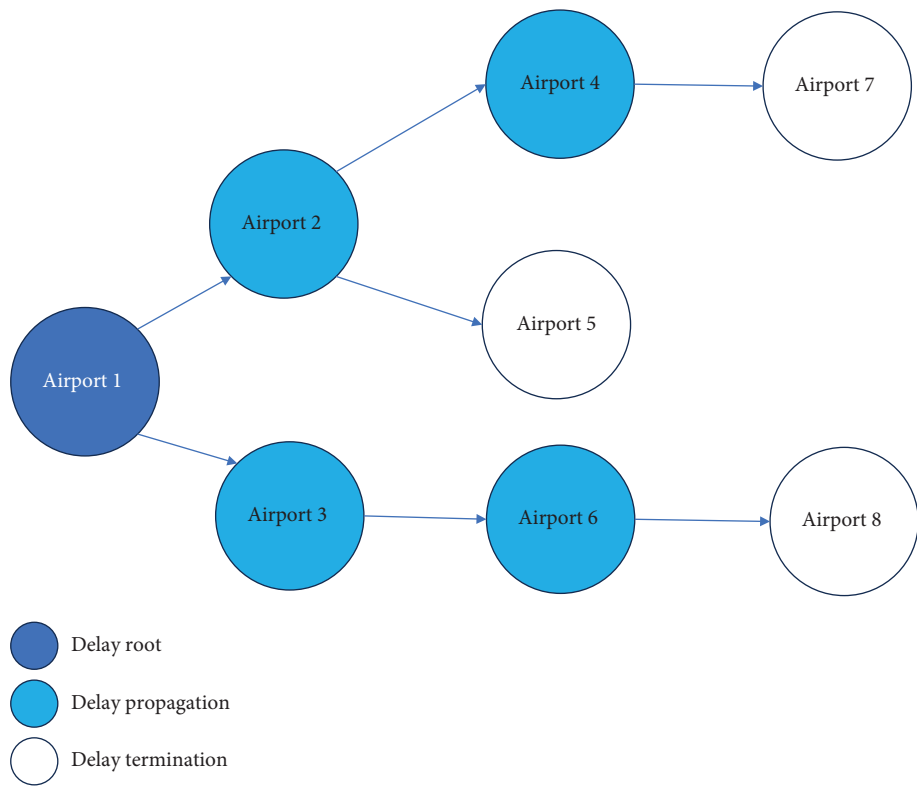


FIGURE 18: Delay propagation tree. The figure is copied from (<https://onlinelibrary.wiley.com/doi/epdf/10.1155/2020/8816615>).

model effectively replicates the distribution of node vulnerability and provides a more accurate ranking of airports compared to the uniform SIS model [96].

3.2.2.2. Delay Propagation Prediction Model Based on Machine Learning. Flight delays pose a significant issue for aviation stakeholders globally. Airlines that experience financial losses and a decline in customer loyalty are particularly impacted [97]. As data storage and computing capabilities advance, data-driven machine learning techniques have become increasingly influential in predicting flight delay propagation. Flight delay prediction primarily consists of two research areas: single-target prediction and full-network prediction, each employing different data-driven machine learning approaches. Naive Bayes [98], DBN, and Random forests [99] are frequently utilized for single-target predictions, particularly in the analysis of individual airports. However, predicting flight delays across the entire network reveals spatial dependencies due to the intricate relationships among regional airports [100]. A delay at one airport can cascade delays throughout the network [101]. Thus, effectively understanding the spatial-temporal correlations at the network level is crucial for enhancing prediction accuracy [102].

Air traffic delay propagation can be divided into two significant links: inter-route transmission (connecting to the airport) and ground transmission within the airport. Some scholars have integrated the multilabel random forest classifier with the approximate delay propagation model to form a chain delay prediction model [103]. The prediction of flight delays on the same aircraft route shows the chain model has predictive ability. The actual departure delay is revised based on the number of times the trip is repeated, which can enhance the model's accuracy. The hybrid prediction method for single-flight arrival and departure delays consists of three components: the arrival delay prediction module, the departure delay prediction module, and the delay propagation module. The structure of this hybrid approach is shown in Figure 19. The delay prediction model serves as the connection between the arrival and departure delay prediction modules. For a specific initial departure delay, repeated execution of the connection module can create a chain model. The other inputs consist of flight schedules and features from the training set. The module for predicting arrival and departure delays utilizes a random forest model to train the chosen features—the delay propagation module is an optimization tool tailored to historical LAAD data. The hybrid prediction approach is shown in Figure 20. The three modules process the actual departure delay of Airport 1 to simulate the departure time of Airport 2. This simulation time can continue to be used in predicting Airport 3 and Airport 4, or adjusted according to the actual departure delay of Airport 2 if there are no new data to support the subsequent prediction, the accuracy is limited. Finally, the overall model prediction is completed only through the arrival delay module of Airport 3 to Airport 4.

This method based on machine learning shows greater accuracy and effectiveness in predicting delays in daily air traffic operations. However, it may face problems such as

high feature dimension and overfitting [104]. As a result, it is essential to enhance the feature selection algorithm to boost the precision and effectiveness of the delay forecasting model.

The machine learning method builds a delay prediction model by preprocessing and analyzing many flight delay data. Although traditional machine learning has achieved great results in predicting flight delays, it requires building a large number of complex feature engineering, which requires a lot of computing resources and time. This method has certain limitations for flight delay prediction with high real-time requirements. It does not consider the adequacy of spatiotemporal feature extraction of flight delay data and lacks the ability to explain the prediction results. Therefore, it encounters bottlenecks in accuracy improvement.

3.2.2.3. LSTM Model Based on Deep Learning. With the deepening of research on delay propagation prediction, deep learning has increasingly played a key role in assessing delay propagation. Major and well-connected airports have higher identifiability in delay feature recognition. The identifiability of delay varies with geographical location and season. In addition, flight datasets exhibit intricate relationships and high-order interactions between spatial and temporal elements, making it challenging to extract salient features and patterns [105]. The deep learning model is effective in evaluating airport delay propagation and extracting spatiotemporal data [106, 107].

Recurrent neural networks (RNNs) [108] and long short-term memory units (LSTMs) [109] have demonstrated extraordinary abilities in capturing temporal dependencies, sparking many studies on using time series models for delay prediction. LSTM is a form of RNN created explicitly for handling sequential data over time [110]. It can analyze aviation sequence data and includes a mechanism to assess the relevance of the information. However, it faces challenges with overfitting when working with limited datasets [111]. The deep LSTM model cannot account for the spatial relationships of spatiotemporal variables. At the same time, bidirectional long short-term memory (BiLSTM) serves as an enhancement of the conventional unidirectional LSTM [112], which can extract spatiotemporal features of flight networks with weather characteristics [113]. BiLSTM combines forward and backward LSTM layers to process data from front to back and from back to front, respectively, to capture past and future information and effectively improve the accuracy of model prediction.

Therefore, some scholars have constructed the TS-BiLSTM-Attention model to predict delays. The BiLSTM unit structure of this model is shown in Figure 21 [114], covering data preprocessing, feature engineering, model training, and effect evaluation. It clearly demonstrates the entire data processing and modeling process from the original data to the final model prediction. This process starts from the original multidimensional time series data, undergoes data preprocessing to ensure data quality, and then extracts statistical features, cross features, and time window features through feature engineering to enhance the input information of the model. The processed

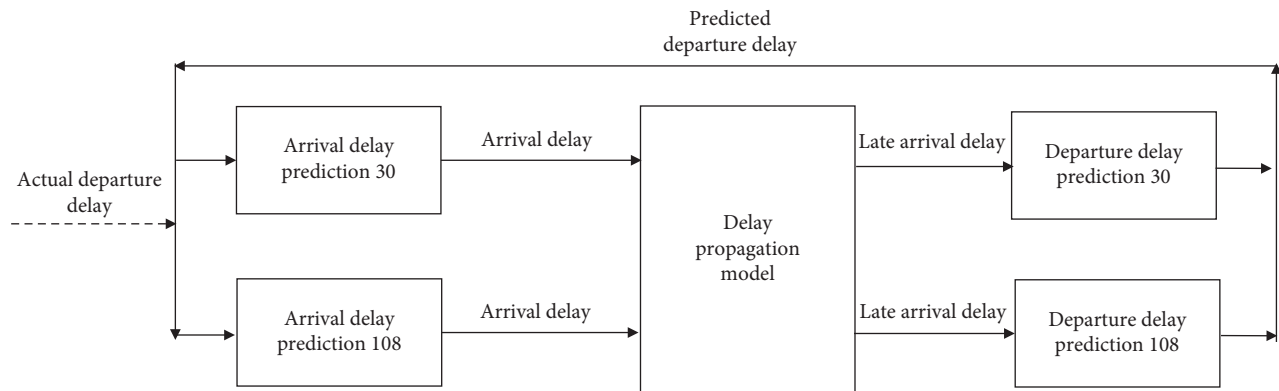


FIGURE 19: Mixed approach for chained delay prediction. The figure is copied from (https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=2490&context=open_access_theses).

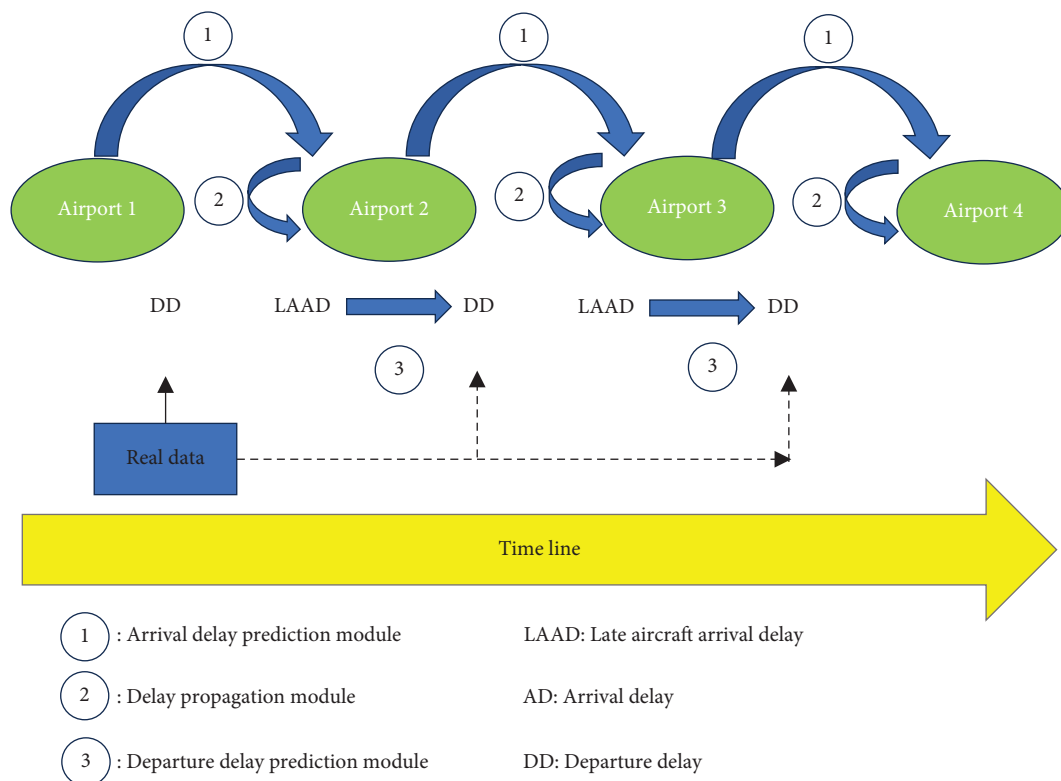


FIGURE 20: An example of a mixed delay prediction approach. The figure is copied from (https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=2490&context=open_access_theses).

data are input into the TS-BiLSTM-Attention model for training, and the model combines the BiLSTM and the Attention mechanism to capture complex spatiotemporal features and output prediction results. Finally, the model is evaluated by root mean square error (RMSE) and mean absolute error (MAE) to verify its prediction performance. The entire process reflects the complete method from data processing to model training and evaluation, highlighting the systematic nature of airport delay prediction based on spatiotemporal features. Comparative verification shows that the TS-BiLSTM-Attention model has high prediction accuracy in most airport clusters, but its prediction

performance is limited in cases with a large number of airports and complex delay propagation situations.

Therefore, it is essential to study the delay prediction between multiple airports from the perspective of the airport network. The researchers integrated the graph convolution embedding module into the sequence-to-sequence LSTM neural network framework, thus developing the airport graph convolution embedding LSTM prediction model. This model can use the historical multiperiod delay data of the airport network to predict the average delay time of future multiperiods and provide high-precision flight delay prediction. Figure 22 shows the DGLSTM network for delay

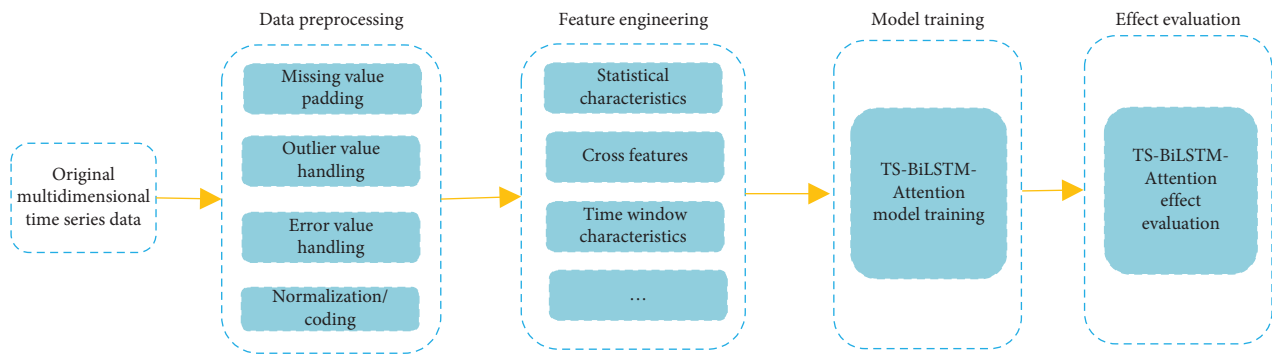


FIGURE 21: Data processing flow of TS-BiLSTM-attention prediction model. The figure is copied from (<https://www.mdpi.com/2226-4310/10/7/580>).

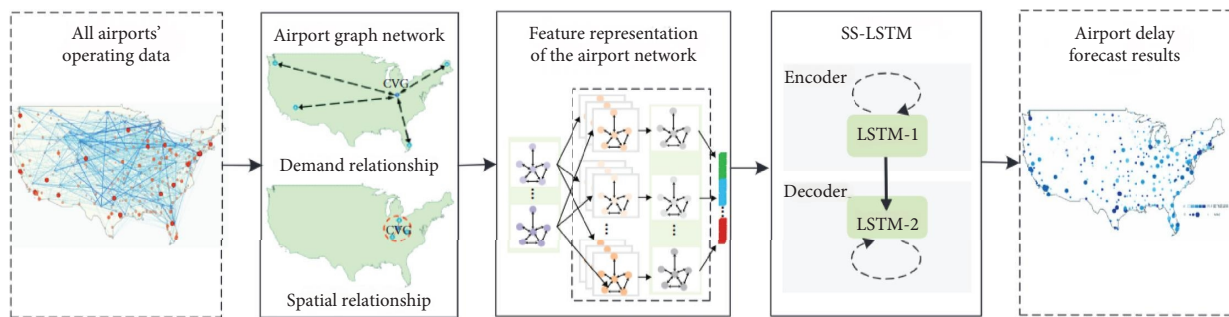


FIGURE 22: Architecture diagram of the DGLSTM airport delay prediction model. The figure is copied from (<https://onlinelibrary.wiley.com/doi/epdf/10.1155/2021/6638130>).

prediction [115]. The network is divided into two stages: encoding and decoding. When encoding, the graph representation is combined with the LSTM module; the decoding is only completed by the LSTM module.

The deep graph-embedded LSTM (DGLSTM) also employs a directed graph network to illustrate airport connections, with edge weights based on spherical distance and flight logs. By incorporating the propagation convolution kernel into the sequence-to-sequence LSTM neural network, the model can recognize the patterns of delay propagation among airports.

Additionally, to accurately analyze the interaction between airports, a model centered on the airport network and a deep time convolution network (DTCN) for predicting airport delays are developed. As shown in Figure 23 [116], the prediction model inputs the delay time series of each airport, outputs the target airport sequence, and introduces the attention mechanism. The research shows that each airport in China's airport network is affected by the other six airports on average and affects the other six airports. Small flights, medium delay airports, and low delay airports are more vulnerable to external influences. This study significantly influences delay management. Furthermore, airports with lower flight volumes and moderate delays tend to have a more significant impact on other airports. In contrast, those with minor delays are more likely to be affected by the operations of different airports. These insights contribute to develop more efficient delay management strategies and enhance the operational effectiveness of the airport network.

In addition, it is necessary to consider the spatial and temporal characteristics in flight delay prediction [117, 118]. Researchers have developed a convolutional LSTM model that integrates spatiotemporal variables such as flight delays, route congestion, airport capacity, and traffic control [119]. The model forecasts the air traffic system congestion by combining spatiotemporal variables with non-spatial time series data. Convolutional long-term and short-term memory network is an end-to-end deep learning framework which combines convolutional neural networks (CNN) and RNNs, making it particularly effective for handling spatiotemporal data. By considering elements such as past flight delays, route congestion, airport throughput, and traffic management, a convolutional long-term and short-term memory network structure is designed to predict future airport flight delay distributions [120].

Deep learning is more and more widely used in flight delays. Deep learning models require a lot of experiments. In the experiment, the parameters are continuously adjusted and the results are optimized. At the same time, the sample quality and model training speed should be considered. If the data quality is not high, it will affect the precision of the forecast outcomes.

3.2.2.4. Spatiotemporal Graph Neural Networks (STGNN). Flight delay propagation prediction technology has undergone significant development, from early statistical regression models and traditional machine learning techniques [121, 122] to more advanced deep learning

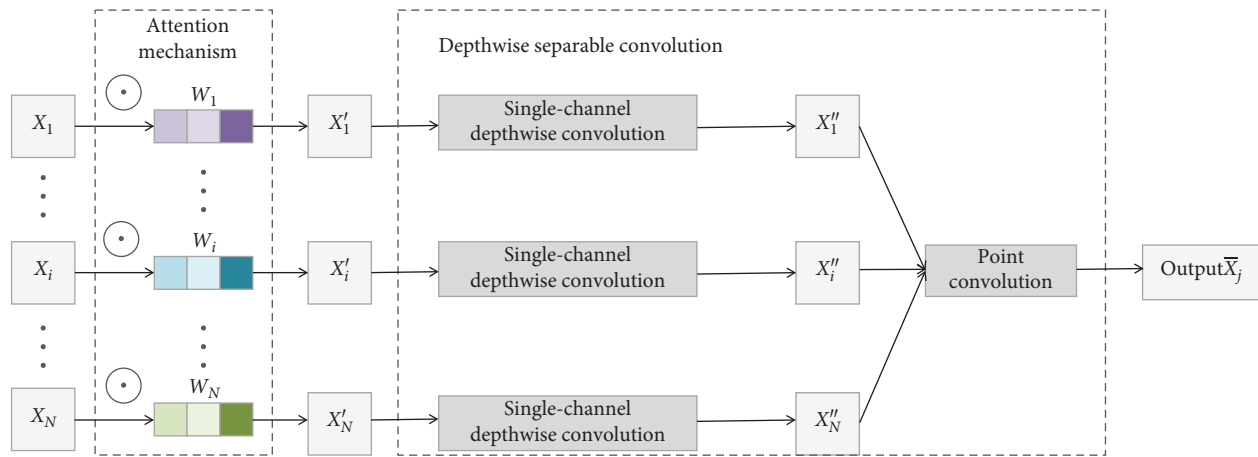


FIGURE 23: The DTCN prediction model architecture based on the attention mechanism. The figure is copied from (<https://www.mdpi.com/2076-3417/12/20/10433>).

models, such as RNN and LSTM networks, which greatly enhance the accuracy of prediction. However, due to the traffic flow transmission between airports, the delay between airports has time and space correlation. As an emerging technology, STGNN effectively integrates spatial correlation and temporal dynamics, provides accurate prediction for flight delay propagation, and shows its versatility in many research fields.

Predicting flight delays in a large airport network is challenging due to the intricate relationships involved. The research on the effective use of the inherent spatial and temporal relationships in the data to predict cross-network flight delays continues to develop. Graph convolutional network (GCN) is good at revealing complex relationships. GCN can model the transient and periodic graph structure information in the airport network and can combine the time convolution block to identify the change pattern of flight delays and reveal the spatial interaction within the network [123]. A deep learning network AG2S-Net model based on an attention mechanism is also applied to delay propagation prediction. The model combines graph convolutional neural network, sequence-to-sequence architecture, and attention mechanism to form a unique framework. As shown in Figure 24 [124], AG2S-Net integrates space, time information, and external variables and flexibly processes variable length sequence data through the encoding-decoding structure. The encoder comprises a graph convolutional neural network, and a BiLSTM neural network effectively uncovers hidden heterogeneous relationships within the network structure data and captures time dependencies in both directions. BiLSTM also analyzes the time dependence of flight delay variables and encodes them as context vectors. The decoder LSTM decodes these vectors to generate multistep predictions. The integrated attention mechanism further enhances the model prediction ability.

Simultaneously, to understand the spatiotemporal patterns of past departure and arrival delays, as well as the influence of external factors like weather conditions on flight delays, some scholars have introduced a spatiotemporal separable GCN known as the spatiotemporal propagation

network (STPN) [125]. The spatiotemporal separable graph convolution is used to model spatial relationships and time dependencies, and the multilayer self-attention layer is used to capture time dependencies and learn complex spatiotemporal dependencies. Figure 25 illustrates the basic building blocks of STPN [126]. The spatially separable multigraph convolution considers the combined spatiotemporal interactions among airports, factoring in geographical closeness, weather factors, and traffic levels that influence delay propagation within the STPN. Additionally, STPN can forecast flight delays several hours in advance. However, the shortcoming is that LSTM performs poorly in dealing with long-term time dependence and is unsuitable for dealing with long-term time dependence in airport flight delays.

To address the long-term periodic graph structure changes over time in airport networks, a Multiscale Spatial-Temporal Adaptive Graph Convolutional Network has been proposed, building on the Graph Convolutional Neural network (GCN) framework. As shown in Figure 26 [127], this system architecture tackles the challenge of predicting flight delays across multiple airports by jointly modeling the spatiotemporal dynamics of the airport network. The model features a deep learning architecture that utilizes graph structure inputs, a time convolution block that leverages the Markov property to capture air traffic time dependencies, and an adaptive graph convolution block designed to learn the complete graph structure input, beneficial for newly established routes during emergencies. The parameterization map and the model convolution parameters are trained and updated collaboratively. The innovation of the model is to analyze the dynamic spatial interaction of the airport network and introduce the adaptive graph convolution learning complete graph structure to improve the short-term and long-term prediction performance. However, there is a problem that lags behind the actual delayed data, which is a common problem in deep learning time series analysis. Early studies have focused on the qualitative impact of flight delays on airport networks, which might not be sufficient for making quantitative forecasts in changing spatial interactions.

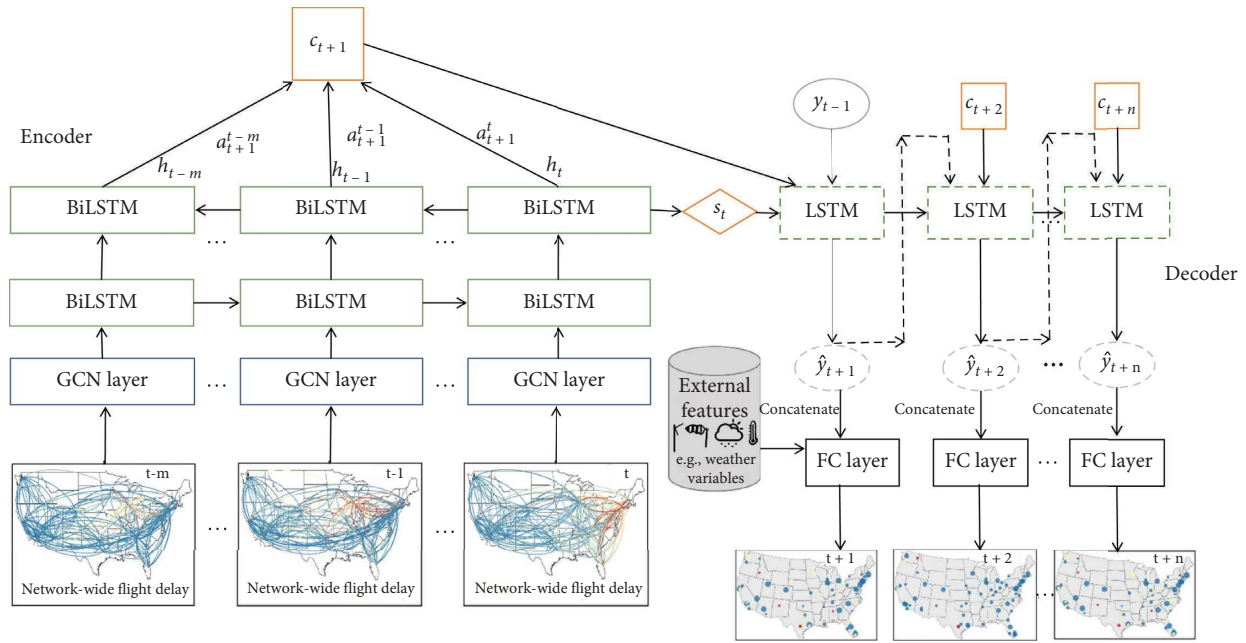


FIGURE 24: The structure of the proposed AG2S-Net. The figure is copied from (<https://www.sciencedirect.com/science/article/pii/S0968090X21003296>).

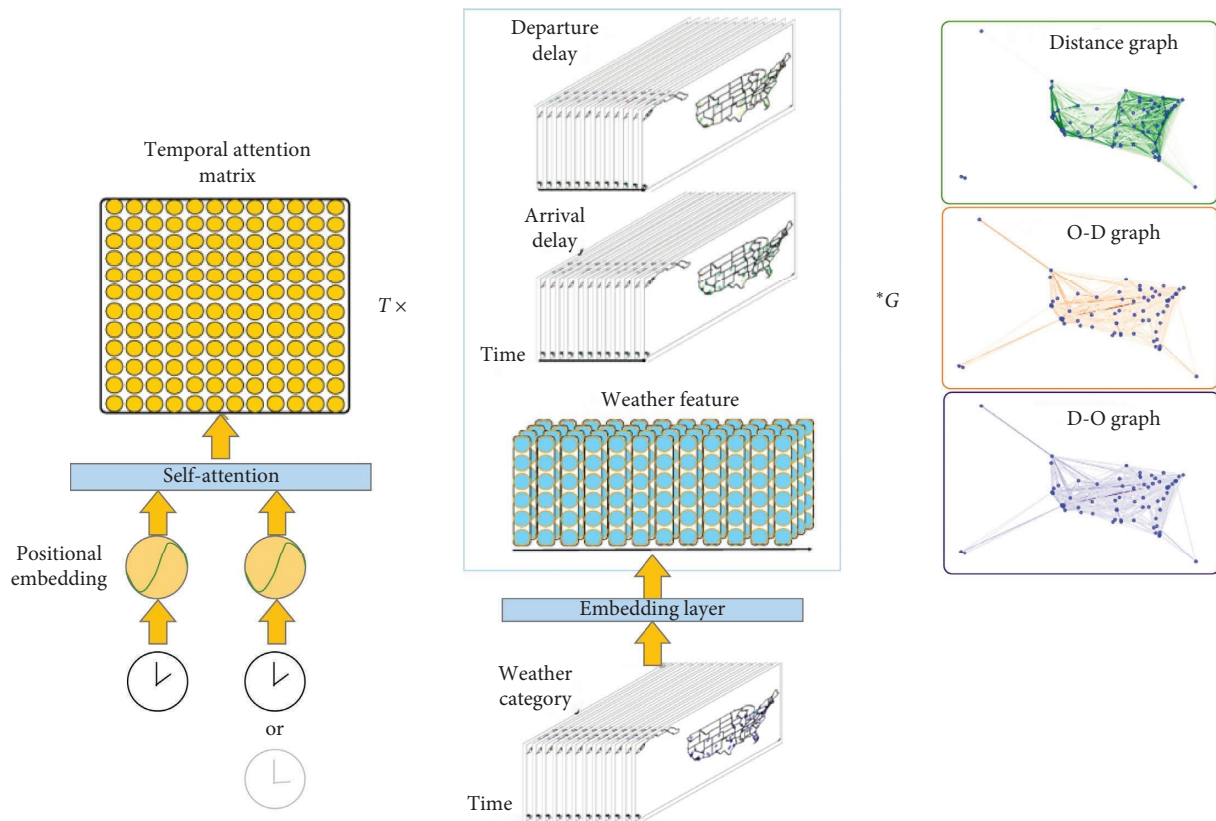


FIGURE 25: Overview of the basic building block (input layer) of STPN. The figure is copied from (<https://arxiv.org/pdf/2207.06959>).

Causal data mining enables researchers to construct and test large-scale causal network structures for air transportation, not only accurately predicting the risk of flight delays but also conducting causal reasoning and

sensitivity analysis [128]. Therefore, some scholars have constructed the second modified transfer entropy-temporal graph convolutional network (SMTE-TGCN) model, which integrates GCN and gated recurrent unit (GRU) to

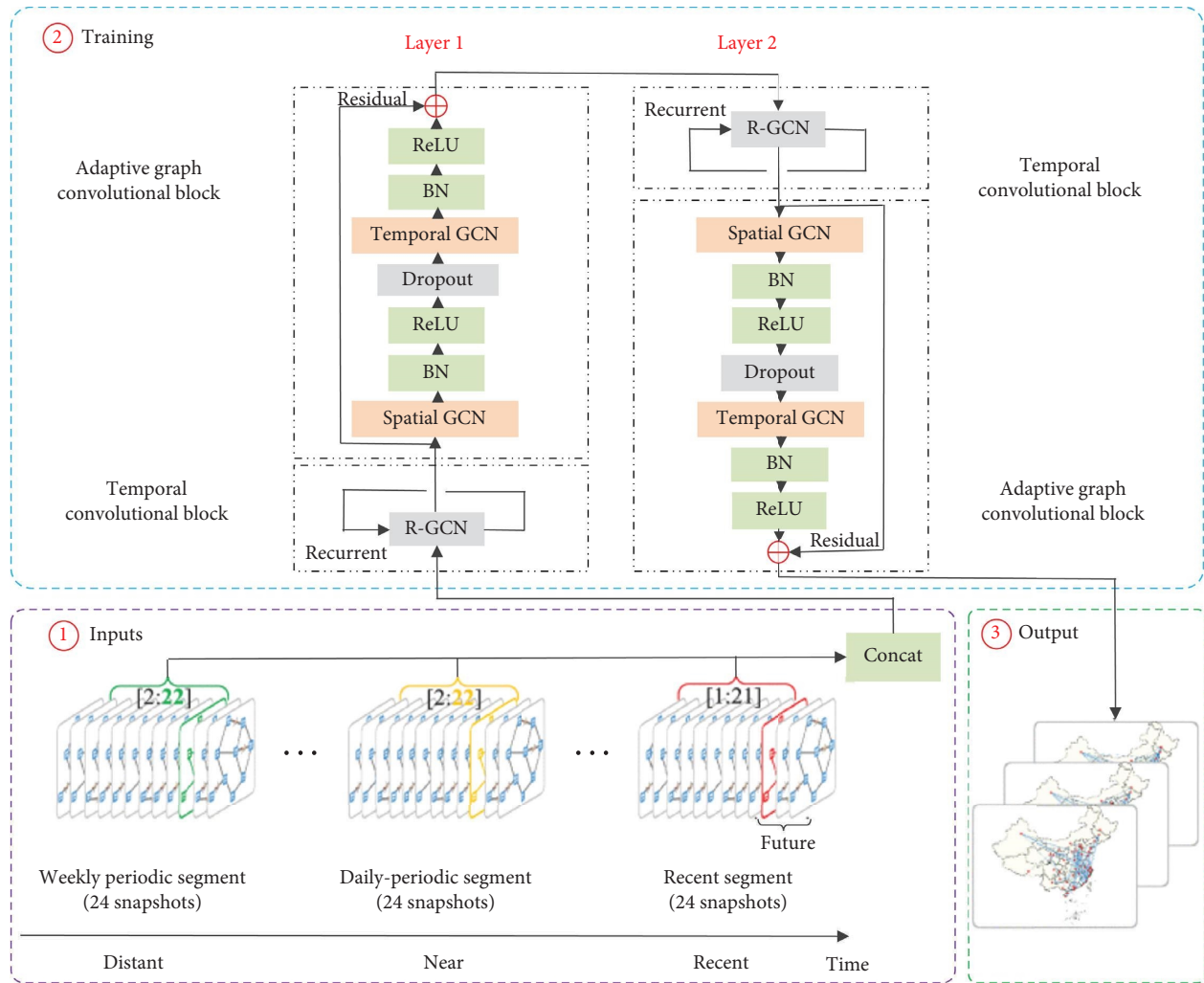


FIGURE 26: System architecture of multiscale spatial-temporal adaptive graph convolutional network. The figure is copied from (<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9512525>).

consider spatiotemporal dependencies, thereby improving prediction accuracy. As shown in Figure 27 [129], this framework consists of three core parts: causal knowledge construction, spatiotemporal feature extraction, and prediction result analysis. In the causal knowledge construction stage, the transfer entropy between airports is calculated and combined with significance tests to construct a causality knowledge-based airport delay propagation network (ADPN). Subsequently, the spatial features of airport nodes are extracted using the GCN, and time features are extracted by combining the GRU to capture the spatiotemporal dependencies of airport delays. Finally, by comparing the prediction results of different models, flight volumes, and delay values, the effectiveness and superiority of the SMTE-TGCN model in airport delay prediction are verified.

Due to the varying capacities and operational management capabilities of different airports, the degree of impact of delay propagation varies among different airports [130, 131]. Traditional GCN modules are unable to extract airport heterogeneity from causal diagrams.

Researchers have proposed a self-corrective STGNN. Its structural framework is shown in Figure 28 [132], including the input layer, training layer, and output layer. In the input layer, the model receives historical flight delay data, geographical information graphs, and causal graphs generated through the Granger causality inference module. The training layer is the core of the model, including the self-correction module, which is used to dynamically adjust the causal graph to improve accuracy; the spatial dependency modeling module, which extracts spatial correlations between airports through GCNs and adaptive weight matrices; and the temporal dependency modeling module, which uses LSTM units (LGRU) to capture long-term dependencies in time series. Finally, the output layer converts the output of the training layer into flight delay prediction values for each airport within the next 1–3 h through a linear layer. The entire architecture integrates spatiotemporal dependency relationships and causality inference to achieve precise prediction of flight delays.

Currently, Granger causality inference has been applied to construct causal relationship diagrams among

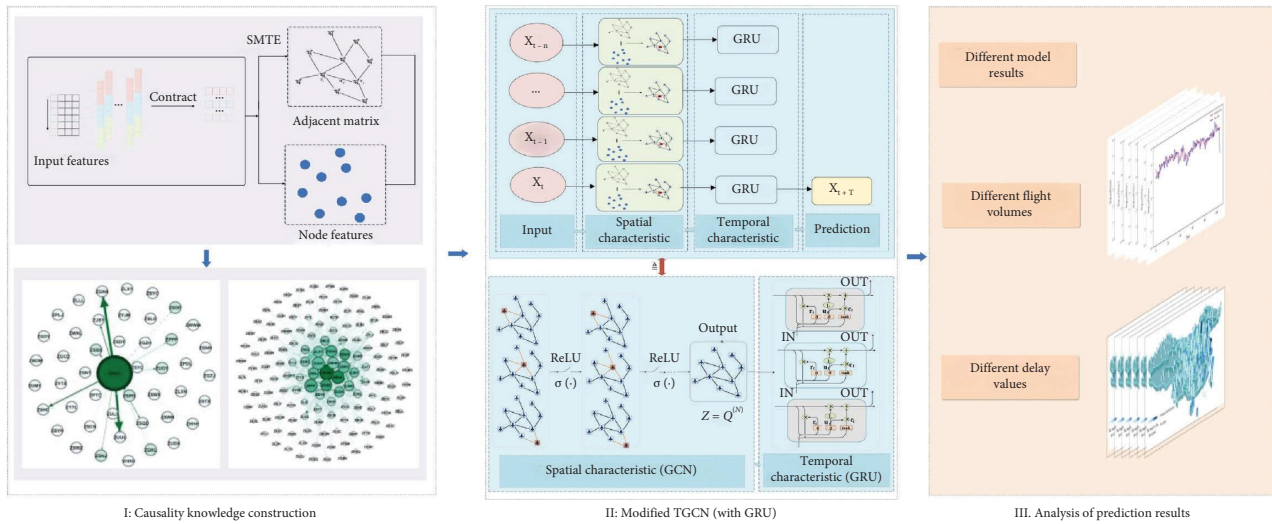


FIGURE 27: Schematic diagram of SMTE-TGCN model framework. The figure is copied from (<https://www.sciencedirect.com/science/article/pii/S0957417423029287#f0005>).

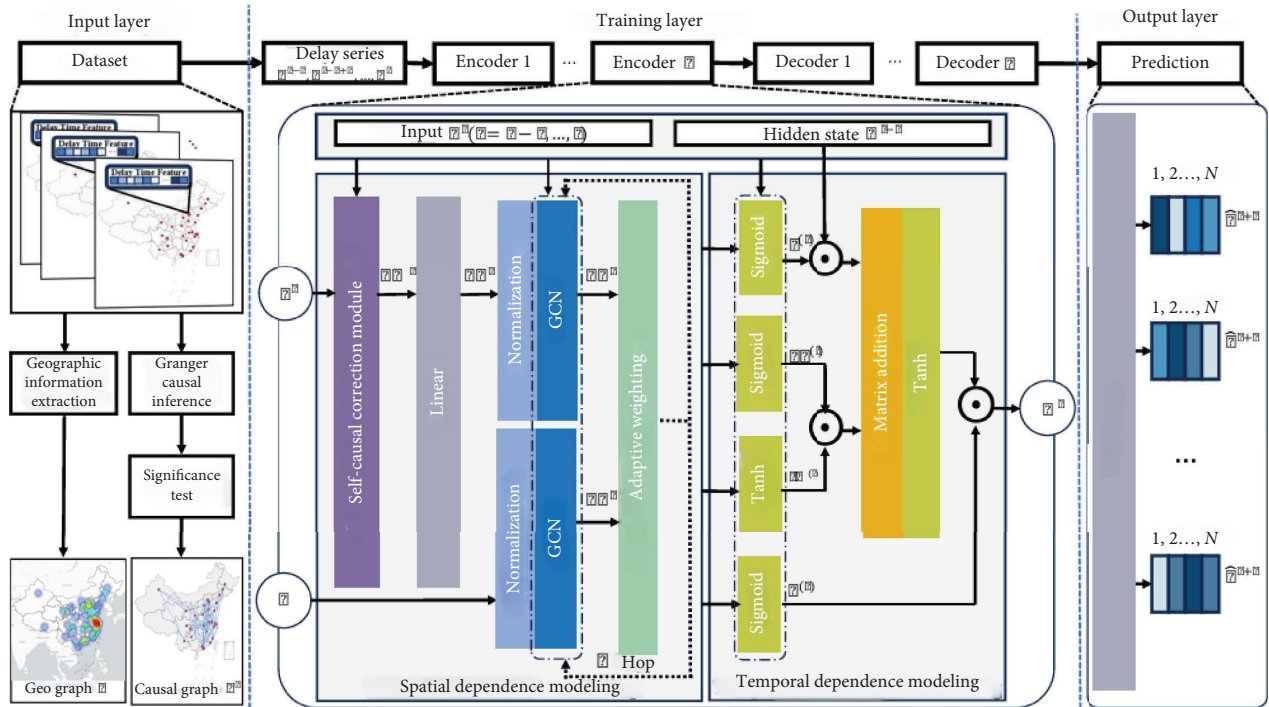


FIGURE 28: The architecture of the self-corrective spatiotemporal graph neural network. The figure is copied from (<https://arxiv.org/pdf/2407.15185>).

multiple airports, while the self-correction module is used to precisely obtain causal relationships between airports, identify the intensity of delay propagation at different time points, and reduce information loss during the construction of causal relationship diagrams. To extract spatial and temporal correlation information more accurately, the heterogeneous information extraction method based on graph convolution is used to model spatial dependence, and the long short-term memory recurrent unit (LGRU) is used to retain long-term memory and extract temporal

dependence. The research results show that large airports are less affected by delay propagation, while small airports are more sensitive. These findings provide important insights for actual flight operations. Moreover, the impact of flight delay propagation may manifest several hours in advance, so when modeling, not only the information of the current time needs to be considered, but also the information of past moments needs to be integrated. How to efficiently aggregate historical information using GCNs has become a current challenge.

3.2.2.5. ATFM. Air traffic flow control plays a key role in improving the efficiency of aviation operations. It helps to avoid delays and safety problems caused by traffic overload. In view of the limited airspace resources, it has become an urgent task to rationally plan flights to meet the growing demand for air transport. In this context, ATFM plays a vital role. It optimizes the take-off time, flight path, and speed of flights by enhancing the efficiency and safety of flight scheduling, aiming to reduce delays and conflicts. The main goal of ATFM is to adjust the flight plan during the flight departure phase to alleviate air traffic congestion [133].

The academic community continues to work on the use of mathematical optimization techniques to address the challenges of ATFM [134, 135]. Early research often defined air traffic scheduling problems as mixed-integer programming problems, where the objective function mainly focused on the total duration of flight delays under capacity constraints. To overcome these challenges, researchers have adopted precise algorithms such as branch and bound [136] and column generation [137]. Although these precise algorithms can provide optimal solutions, their computational costs are often difficult to meet the needs of large-scale applications in the real world. In contrast, the metaheuristic algorithm has shown their potential as a strategy for finding approximate optimal solutions within a finite time [138, 139]. The modeling of ATFM problems requires consideration of multiobjective optimization, as it can provide decision-makers with more flexibility and choice space, especially in complex ATFM scenarios that require a balance between efficiency and safety [140]. Therefore, the recent research trend is to extend the model to include multiple objectives [141, 142]. In the field of ATFM, multiobjective optimization problems have demonstrated their unique value. It not only includes multiple core principles and competing objectives, but also provides decision-makers with broader flexibility and decision-making options in practical operations, especially when making difficult choices between efficiency (such as delay) and safety [143].

In order to address the shortcomings of existing research, researchers have developed a new model for multiobjective ATFM problems, which covers comprehensive ATFM operations such as ground and air delays, route adjustments, and control of flight speed [144]. This new model enhances the adaptability and flexibility of ATFM by allowing speed adjustment during flight. The article first introduces an innovative multiobjective ATFM problem model that supports speed adjustment in the four-dimensional trajectory of flight to enhance management flexibility and operability. Subsequently, researchers develop a novel multiobjective evolutionary algorithm, which features individual encoding and search strategies designed for specific problems, including multigene encoding schemes and composite crossover operators. In order to strike a balance between exploration and development, a population clustering technique based on elbow points is proposed, which divides the population into elbow point individuals, nondominant individuals, and dominant individuals, and applies different mutation strategies. By constructing test cases of different scales and using actual air

traffic data for extensive experimental verification, it is found that compared with the existing ATFM operations, the new algorithm can reduce conflict events and significantly reduce overall delays.

In addition to the above research, a coevolutionary strategy specifically for large-scale multiobjective ATFM problems has been developed [145]. The researchers first construct an innovative multiobjective coevolutionary framework to reduce flight delays and conflicts while ensuring aviation safety. The framework uses the elbow solution in the external repository to strengthen the collaboration between subgroups. Subsequently, an innovative fuzzy partitioning technique is proposed to split the large-scale ATFM problem into smaller subproblems based on the temporal and spatial correlation of the aircraft. In addition, the researchers also designed a contribution-based random resource allocation mechanism to allocate computing resources for subproblems with uneven load in an automated manner. Experimental results show that this method can significantly improve performance and reduce flight delays and conflicts compared with other algorithms within the same number of fitness evaluations.

ATFM is crucial for improving aviation efficiency. It coordinates the growing demand for air traffic with limited airspace resources through rational planning of aircraft flight plans, which is crucial for air navigation service providers to ensure the safety and sustainability of air transport. With the surge in air traffic, the number of scheduled flights is also rapidly increasing, which puts higher demands on the efficiency of flight scheduling. Therefore, in-depth research on ATFM is of great significance in reducing flight delays and mitigating the impact of delay propagation.

4. Conclusions

The issue of flight delay propagation is a significant factor hindering the growth of the civil aviation sector, significantly affecting both airlines and passengers. Therefore, it is essential to predict the propagation delay accurately. This can not only enable passengers to make reasonable schedules but also allow airports, airlines, and airspace management personnel to coordinate, so as to take positive measures to minimize pressure and loss. Based on the latest studies regarding the prediction of flight delay propagation, the following conclusions can be made:

1. At present, research on predicting flight delay propagation mainly focuses on two aspects: delay propagation within airlines and delay propagation between airports. In the study of flight delay propagation mechanisms within airlines, economic models, Bayesian networks, function models, and propagation trees are mainly used to analyze propagation characteristics. However, due to the complexity of flight delay propagation between airports, these models have certain limitations in analyzing the mechanism of delay propagation between airports. More researchers use time Petri nets, Cox proportional

hazards models, queuing networks, and complex network theory to analyze propagation characteristics. Research shows that there are significant differences between the delay propagation mechanisms within airlines and between airports. The propagation of delays within airlines is mainly influenced by the interconnectedness of flight chains, such as aircraft allocation, crew shifts, and tight connections to flight schedules, which can lead to delays in one flight quickly spreading to subsequent flights through resource sharing and time dependence. The propagation of delays between airports is more driven by external factors such as weather, route congestion, and flight traffic management, and its propagation path is usually related to the connectivity of the airport network and the concentration of flight traffic. Therefore, internal delay propagation within airlines is dominated by internal operational efficiency, while interairport delay propagation is more dependent on regional network structure and external environmental conditions.

2. In terms of predicting flight delay propagation, traditional machine learning models, dynamic data-driven models, and simulation models perform well in predicting internal delays within airlines, but they have certain limitations in dealing with complex and changing propagation environments between airports. The prediction of delay propagation between airports mainly uses machine learning, deep learning, spatiotemporal modeling methods, and ATFM combined with flight scheduling, historical data, and external variables, to predict the dynamic evolution of delays in internal flight chains and cross airport networks, providing theoretical and practical support for optimizing scheduling and improving operational robustness.
3. Machine learning algorithms can analyze historical flight data and identify patterns and trends to predict future delays. However, flight delays are affected by multiple factors involving complex flight and meteorological data. Traditional machine learning models need high-quality and sufficient data support to ensure accuracy and robustness, and the data processing process is more complicated. Deep learning based on traditional machine learning techniques is more suitable for large datasets and high-dimensional feature data. With the continuous updating of datasets, the model will be constantly updated. Furthermore, by utilizing its network for training, it can thoroughly analyze and extract features from the dataset, examine the relationships among the data, and enhance the precision of flight delay forecasts. However, the irregular distribution of airports within the network and the varying levels of influence among them make it challenging to convert the data into a uniform grid format. Convolutional neural networks are only suitable for structured data such as images. The graph convolutional neural network is capable of directly extracting features from graph-structured data, making it more appropriate for predicting the propagation of flight delays.
4. Most existing studies on flight delays are carried out from the time and rarely involve the diversity of time and space. In the study of flight delay timing prediction, it has been found that there are multidimensional influencing factors. If its spatial attributes are included, it will inevitably bring more complex data. Compared with other methods, deep learning and spatiotemporal network modeling are more suitable for research. Moreover, due to the strong correlation between different airports, flight delays have a significant propagation effect. Delay is not only affected by the airport where the flight takes off and lands, but also by other airport factors. This type of correlation between airports requires a large amount of flight data and weather data as model training samples, which can easily lead to dimensionality explosion problems, and preliminary manual feature selection has a great impact on the training effect of the model. Although machine learning and artificial intelligence technologies have powerful data processing and analysis capabilities, they also require continuous optimization and improvement of algorithms to enhance the accuracy and reliability of predictions.
5. Missing data and uncertainty are still a difficult problem in flight delay prediction. Real-time data update and dynamic analysis require efficient data processing and analysis capabilities, which puts forward higher requirements for technical implementation and system performance. How to obtain more accurate and comprehensive data, especially real-time data, is the key to improve the accuracy of prediction. In addition, the uncertainty of external factors such as weather and traffic conditions increases the difficulty of prediction.
6. By understanding the propagation mechanism of delays, flight scheduling can be optimized, airline operational efficiency can be improved, and economic losses caused by delays can be reduced. Meanwhile, research on delay propagation provides scientific decision support for airports and air traffic control departments, which helps improve flight flow management and resource allocation. In addition, with the application of technologies such as big data and artificial intelligence, research on delay propagation can achieve more accurate real-time prediction and dynamic adjustment, laying the foundation for building a more intelligent and reliable air transport system, thereby enhancing passenger experience and promoting the sustainable development of the aviation industry.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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